Random Projection-Based Sub-Pixel Target Detection for Hyperspectral Image With t-Distribution Background

Qingke Zou, Jie Zhou, Yubo Ma, and Mingjie Luo

***Abstract*— Sub-pixel target detection is a challenging task in hyperspectral image processing. Most statistical detectors rely on the estimation of the background covariance matrix. In contrast to the traditional approaches by adopting global covariance matrix estimation from all image pixels, the local estimation within image segments can significantly improve detection performance for complex backgrounds in many scenarios. However, the local covariance matrix estimate may be unstable due to the high spectral dimension of the hyperspectral image, especially when the size of the local sample representing background pixels, is relatively small. In this work, a random projection (RP) is employed to reduce the spectral dimension, and a spectral similarity-based dual-window (SSDW) strategy is suggested to appropriately extract the local background statistical properties. So, a kind of sub-pixel target detector is developed in a statistical hypothesis testing framework for different observation models under t-distribution background. Especially, the projection dimension is determined without any extra experiment or training and ensures asymptotic optimality of detection power under some conditions. The superior performance of the proposed detectors is demonstrated by some synthetic data and real hyperspectral images.**

***Index Terms*— Generalized likelihood ratio tests (GLRTs), hyperspectral image, random projection (RP), spectral similarity, sub-pixel target detection.**

I. Introduction

HYPERSPECTRAL images are "data cubes" consisting spatial information and spectral information of scene objects. This kind of data cube provides great convenience for distinguishing among different objects and detecting the specified targets [1], [2], [3], [4], [5], [6], [7]. Consequently, hyperspectral remote sensing technology has developed rapidly and is gradually becoming one of the cutting-edge technologies for military and civilian needs. Notable uses include the detection of man-made materials and aircraft in military reconnaissance, examination of mineral and geological formations in resource prospecting, and spectral analysis of soil in agriculture [8], [9], [10], [11], [12], [13]. Driven by diverse application demands, hyperspectral image detection algorithms have seen considerable advancements (see [14], [15], [16] and references therein). In particular, recent years have witnessed remarkable advances in machine learning-based approaches (see an overview [17]). Depending on whether the spectral information of the target is pre-given or not, these algorithms can be divided into two categories: target detection and anomaly detection. The former focuses on locating a target of interest. However, the relatively low spatial resolution of hyperspectral sensors, coupled with the intricate spread of terrain and objects, results in scant instances of pure pixel elements in imagery. Typically, the target only reflects a proportion of one pixel, known as a sub-pixel target [18], [19], [20].

The burgeoning interest in hyperspectral target detection has accelerated algorithmic research, yielding plenty of statistical test-based methodologies. These approaches formulate target detection as a binary hypothesis: target absence (H_0) or target presence (H_1) [21], [22], [23], [24]. Hypothesis testing-derived algorithms, including those delineated in [25], [26], and [27], trace their origins to the classical likelihood ratio test (LRT) and its derivative, generalized LRT (GLRT). An important reason is that statistical testing theory provides a guarantee of the optimality of LRT-based decisions in certain situations.

The target detection algorithms are categorized into full-pixel and sub-pixel detection depending on whether the target covers the full or portion of the observation in alternative hypothesis [28]. As a classical full-pixel target detector, the matched filter (MF) models the two competing hypotheses by a multivariate Gaussian distribution with equal variance [28]. The MF is designed by maximizing the signal-to-noise ratio of the output. However, its efficacy is contingent on prior knowledge of the target mean, which is usually not readily available. To circumvent this limitation, the adaptive MF (AMF) integrates a prior target spectrum into the hypothesis framework, enhancing its applicability [26].

Depending on whether additive noise is included in the background when modeling the background, as [28], the sub-pixel target detection algorithms can be divided into structured and unstructured detectors.

Structured detectors portray the background via a subspace model. There is a widely used class of structured detectors that focus on finding an optimal projection matrix.

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By projecting data into a lower-dimensional space, these algorithms aim to suppress background and accentuate the target signal, effectively widening the gap between target and non-target. The orthogonal subspace projection (OSP) [29], initially developed for dimensionality reduction, exemplifies this category and is extensively applied in target detection. It leverages a linear mixed model (LMM) to represent mixed pixels as amalgams of target and background spectra. The typical representatives of such algorithms include kernel OSP (KOSP) [30], target-constrained interference-minimized filter (TCIMF) [31], kernel TCIMF (KTCIMF) [32], constrained energy minimization (CEM) [33], hierarchical CEM (HCEM) [34], ensemble CEM (ECEM) [35], and kernel CEM (KCEM) [36]. It is a remarkable fact that the above detectors do not make any assumption on the statistical distribution of the noises. The adaptive matched subspace detector (AMSD) [37], another structured approach using the LMM, has been noted for its exceptional target detection performance. However, the treatment of correlation matrix eigenvectors as endmembers in the design of AMSD does not adhere to non-negativity and uniqueness, thus leading to a loss of physical interpretability of this algorithm. To address this issue, some hybrid detectors have been introduced [38], merging physical significance with background modeling.

The unstructured detectors assume that the background contains additive noise, an example of which is the adaptive cosine/coherent estimator (ACE) [39]. It provides a uniform most powerful (UMP) invariant test statistic through the GLRT under the assumption that the background obeys a multivariate normal distribution. Moreover, the non-Gaussian nature of hyperspectral data demonstrated in [40] and [41] prompts some extensions of the AMF to be elliptically distributed under the additive model, e.g., the elliptical-contour GLRT (EC-GLRT) in [42]. Afterward, the GLRTs corresponding to the additive model, replacement model, and mixed model are derived under the background characterized by matrix-variate t-distributions [43]. These adaptations have verified that the GLRTs under t-distribution background are consistent with their Gaussian counterparts.

Most detection algorithms aim at additive models and rarely employ replacement models. One of the detectors based on replacement models is the well-known finite target MF (FTMF), which is the replacement-model version of the AMF under the Gaussian distribution background [28]. Compared to AMF, it takes into account the abundance of targets, thus reducing false alarms caused by background replacing part of the target. An extension of FTMF to accommodate elliptically contoured multivariate t-distribution, known as the EC-FTMF, provides a closed-form solution of the optimized replacement model [44].

One of the inherent challenges in hyperspectral image target detection is that the statistical properties of the background (e.g., mean and covariance) are not known exactly. It is desired to select a sample uncorrelated with the target of interest for extracting the statistical information of the background. The selection of such a sample is crucial, as it significantly influences detection performance. Depending on whether the sample comes from the whole image or some near pixels under test, the detection is classified into global and local approaches. The traditional approaches select all the image pixels to extract the background statistical information [39], [45]. However, in most cases, significantly superior performance has been reported when using some spectral similarity-based techniques, like histogram segmentation, to select a sample in the local region including the pixel under test [46], [47].

Hyperspectral data is inherently rich in spectral information, aiding significantly in target detection. However, it comprises numerous narrow bands, leading to spatial, inter-spectral, and band-to-band correlations, which in turn produce considerable redundant data and potentially misleading information. Fortunately, it has shown that the redundancy can be removed by some techniques such as dimensionality reduction [48]. Random projection (RP), in particular, excels in reducing dimensions while maintaining data integrity and does not necessitate training data, having gained widespread use in image processing [49]. It can make better utilization of the structural information of covariance matrix [50], and overcome the drawback that the covariance matrix estimate may be unstable due to high spectral dimensionality [51], especially when the size of the sample representing the background is relatively small in local detection.

Motivated by the aforementioned observations on hyperspectral image detection, this work is devoted to providing effective test statistics for local sub-pixel target detection by exploiting joint spatial–spectral information. The contributions of this work mainly include the following.

1) The hypothesis test statistics for different sub-pixel target detection models are derived separately based on the RP method for t-distribution background. The proposed procedure avoids the unstable estimation of the covariance matrix when the sample size is smaller than the spectral dimension by testing the projected sample in a low-dimensional spectral space, and reduces the computational complexity due to the dimension-reduced covariance matrix.
2) An efficient method for determining the projection dimension is introduced. It significantly improves the availability of dimensionality reduction since any extra training is unnecessary. More importantly, the suggested dimension has asymptotic optimality in the sense of maximizing detection power under some conditions.
3) A spectral similarity-based dual-window (SSDW) strategy is proposed to more accurately select samples for extracting background information. Compared with the general dual window (GDW) strategy, the SSDW strategy can effectively improve the detection performance.

The rest of this article is organized as follows. In Section II, the sub-pixel target detection problem is formulated as a statistical hypothesis testing with different observation models, and then the corresponding generalized likelihood ratio (GLR) is derived via RP and SSDW strategy. Section III provides some examples for evaluating the performance of the proposed detectors on real hyperspectral images and synthetic data. Section IV gives conclusion. Proofs of theoretical results are given in Appendix .

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II. RP Method

A. GLRs for Three Models

The spectral reflectances of the pixel under test and the background are denoted by $\mathbf{y} \in \mathbb{R}^d$ and $\mathbf{z} \in \mathbb{R}^d$ (d is the spectral dimension), respectively. Our task is to determine whether $\mathbf{y}$ contains a signature of interest $\mathbf{t} \in \mathbb{R}^d$ or exclusively contains the background $\mathbf{z}$ that is uncorrelated with the target $\mathbf{t}$. Since the statistical information of the background is not fully known, we select the sample $\mathbf{z}_i \in \mathbb{R}^d$ ($i = 1, \ldots, n$) from around the pixel under test $\mathbf{y}$ to portray it, and this sample is assumed to be identically distributed with the background $\mathbf{z}$. Furthermore, we can formulate the detection problem as a classical statistical hypothesis test. Let H_0 be the target absence and H_1 be the target $\mathbf{t}$ presence. Based on the LMM, the detection of the target can be formulated as testing the following statistical hypotheses:

$$\begin{aligned} H_0 &: \mathbf{y} = \mathbf{z}; \quad \mathbf{z}_i \overset{D}{=} \mathbf{z}, i = 1, \ldots, n, \\ H_1 &: \mathbf{y} = \alpha\mathbf{t} + \beta\mathbf{z}; \quad \mathbf{z}_i \overset{D}{=} \mathbf{z}, i = 1, \ldots, n \end{aligned} \tag{1}$$

where $\overset{D}{=}$ means "has the same distribution as," $\alpha \in (0, 1)$ represents the unknown abundance of the target $\mathbf{t}$, and $\beta > 0$ is the unknown scale factor of the background. Depending on β, as [43] and [52], three different observation models in the hypothesis H_1 are considered as follows.

1) Additive model: $\beta = 1$.
2) Replacement model: $\beta = 1 - \alpha$.
3) Mixed model: β is a positive real number.

Note that the mixed model degenerates into the additive or replacement model if $\beta = 1$ or $1 - \alpha$ occurs exactly.

The selected sample for extracting background information is denoted as $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n] \in \mathbb{R}^{d\times n}$. As in [43], the pixel under test and pixels in background $[\mathbf{y}, \mathbf{Z}] \in \mathbb{R}^{d\times(n+1)}$ are formulated as a matrix-variate t-distribution $\mathcal{T}_{d,n+1}$ with ν degrees of freedom. Suppose that the mean vector and covariance matrix of the background are $\boldsymbol{\mu} \in \mathbb{R}^d$ and positive definite $\Sigma \in \mathbb{R}^{d\times d}$, respectively. Then the binary hypotheses (1) can be rewritten as

$$\begin{aligned} H_0 &: [\mathbf{y}, \mathbf{Z}] = \mathcal{T}_{d,n+1}(\nu, \mathbf{M}_0, (\nu-2)\Sigma, \mathbf{I}_{n+1}), \\ H_1 &: [\mathbf{y}, \mathbf{Z}] = \mathcal{T}_{d,n+1}\left(\nu, \mathbf{M}_1, (\nu-2)\Sigma, \begin{pmatrix} \beta^2 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & \mathbf{I}_n \end{pmatrix}\right) \end{aligned} \tag{2}$$

where $\mathbf{M}_0 = [\boldsymbol{\mu}, \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}]$ and $\mathbf{M}_1 = [\alpha\mathbf{t}+\beta\boldsymbol{\mu}, \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}]$ with $\mathbf{1}_n$ being the $n \times 1$ vector having all entries 1, and $\mathbf{I}_p$ denotes the $p \times p$ identity matrix.

Most detectors require the sample covariance matrix to be invertible. However, it is quite common that the spectral dimension is larger than the sample size in local detection. In this situation, the sample covariance matrix is singular, which results in the failure of these detectors to work. To avoid this drawback while removing redundant information from hyperspectral images, we will adopt the RP in the statistical hypothesis model and construct the RP-based test statistics.

For an integer $k \in \{1, \ldots, \min\{n, d\}\}$, let $\mathbf{P} \in \mathbb{R}^{k\times d}$ be a random matrix with independent and identically distributed (i.i.d.) entries sampled from the standard normal distribution $N(0, 1)$, and be independent of the data $[\mathbf{y}, \mathbf{Z}]$. Note that $\mathbf{P}$ is row-full rank with probability 1. Assume that $\mathbf{Z}$ is uncorrelated with the target $\mathbf{t}$. Conditioning on a given matrix $\mathbf{P}$, $[\mathbf{Py}, \mathbf{PZ}]$ follows a matrix-variate t-distribution.

In this work, based on the RP matrix $\mathbf{P}$, we consider the following statistical hypothesis testing:

$$\begin{aligned} H_{0,\mathrm{proj}} &: \mathbf{Py} = \mathbf{Pz}; \quad \mathbf{z}_i \overset{D}{=} \mathbf{z}, i = 1, \ldots, n, \\ H_{1,\mathrm{proj}} &: \mathbf{Py} = \alpha\mathbf{Pt} + \beta\mathbf{Pz}; \quad \mathbf{z}_i \overset{D}{=} \mathbf{z}, i = 1, \ldots, n. \end{aligned} \tag{3}$$

Theorem 1: The GLR for the binary hypothesis test problem (3) is given as follows.

1) For the additive model

$$R_{\mathrm{PA}} = \frac{\left(\frac{n}{n+1}\left(\mathbf{P}\check{\mathbf{y}}\right)^{\mathrm{T}}\tilde{\mathbf{S}}^2\tilde{\mathbf{t}}\right)^{n+1}}{\left(\left(1+\frac{n}{n+1}\left(\mathbf{P}\check{\mathbf{y}}\right)^{\mathrm{T}}\tilde{\mathbf{S}}^2\left(\mathbf{P}\check{\mathbf{y}}\right)\right)\left(\tilde{\mathbf{t}}^{\mathrm{T}}\tilde{\mathbf{S}}^2\tilde{\mathbf{t}}\right)\right)^{\frac{n+1}{2}}} \tag{4}$$

with $\check{\mathbf{y}} = \mathbf{y} - \bar{\mathbf{z}}$, $\tilde{\mathbf{S}} = (\mathbf{PSP}^{\mathrm{T}})^{-1/2}$, $\mathbf{S} = \mathbf{ZZ}^{\mathrm{T}} - n\bar{\mathbf{z}}\bar{\mathbf{z}}^{\mathrm{T}}$, $\bar{\mathbf{z}} = \mathbf{Z}\mathbf{1}_n/n$ and $\tilde{\mathbf{t}} = \mathbf{Pt}$.

2) For the replacement model

$$R_{\mathrm{PR}} = \frac{\left(1+\frac{n}{n+1}\|\tilde{\mathbf{S}}\mathbf{P}\check{\mathbf{y}}\|^2\right)^{\frac{n+1}{2}}}{\min_\alpha(1-\alpha)^k\left(1+\frac{n}{n+1}\frac{\|\tilde{\mathbf{S}}\mathbf{P}\bar{\mathbf{y}}\|^2}{(1-\alpha)^2}\right)^{\frac{n+1}{2}}} \tag{5}$$

where $\bar{\mathbf{y}} = \mathbf{y} - \alpha\mathbf{t} - (1-\alpha)\bar{\mathbf{z}}$, and $\|\cdot\|$ denotes the Euclidean norm of vector.

3) For the mixed model

$$R_{\mathrm{PM}} = \frac{\left(1+\frac{n}{n+1}\|\tilde{\mathbf{S}}\mathbf{P}\check{\mathbf{y}}\|^2\right)^{\frac{n+1}{2}}}{\min_\beta \beta^k\left(1+\frac{n}{n+1}\frac{\|\mathbf{P}^{\perp}_{\tilde{\mathbf{S}}\tilde{\mathbf{t}}}\tilde{\mathbf{S}}\mathbf{P}(\mathbf{y}-\beta\bar{\mathbf{z}})\|^2}{\beta^2}\right)^{\frac{n+1}{2}}} \tag{6}$$

with $\mathbf{P}^{\perp}_{\tilde{\mathbf{S}}\tilde{\mathbf{t}}} = \mathbf{I}_k - (\tilde{\mathbf{S}}\tilde{\mathbf{t}})(\tilde{\mathbf{S}}\tilde{\mathbf{t}})^{\mathrm{T}}/\|\tilde{\mathbf{S}}\tilde{\mathbf{t}}\|^2$.

Proof: See Appendix A. □

Remark 1: The matrix $\tilde{\mathbf{S}}$ in the GLRs (4)–(6) is well-defined. In fact, conditioning on a given $\mathbf{P}$, $(\mathbf{Pz}_1, \mathbf{Pz}_2, \ldots, \mathbf{Pz}_n)$ is a simple random sample with size n from some k-dimensional population, then, under a weak condition [53], its sample covariance matrix $(1/n)\mathbf{PSP}^{\mathrm{T}}$ is non-singular with probability 1 owing to $k \le \min\{n, d\}$.

For the GLR (5), solving the parameter α in the optimization problem can be translated into finding the unique positive root of a univariate quadratic function [54]. Similarly, for the GLR (6), the solution of the parameter β in the optimization problem is determined by the unique positive root of a second-order polynomial equation [55].

B. Selection of Projection Dimension

The selection of projection dimension earns great concern, which directly determines the effectiveness of dimensionality reduction. Traditional methods often determine the projection dimension through extensive training of datasets. Depending on the task, different algorithms are generated. For example, the projection dimension is selected by minimizing the structural error in principal component analysis (PCA), and by maximizing the category separability in linear discriminant

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analysis (LDA). These algorithms, albeit effective, are computationally intensive and may compromise the efficiency of the projection. In this work, we propose a direct approach for setting the projection dimension, predicated on the size of the sample. This method aims to minimize computational overhead in image detection tasks without compromising the integrity of the dimensionality reduction.

The choice of projection dimension is intimately related to the GLR. In this work, we set the optimal oracle projection dimension when α and β have not been determined or estimated.

Under the assumption of $\mathbf{y}$ and $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n$ being i.i.d. as [24], in the proof of Theorem 1, the maxima of likelihood functions under the null and alternative hypotheses, respectively, have been proved as [see (32) and (33)]

$$L_0(\mathbf{Py}, \mathbf{PZ}) = C|\mathbf{P}[\mathbf{y}, \mathbf{Z}]\mathbf{P}_{n+1}^{\perp}[\mathbf{y}, \mathbf{Z}]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}|^{-\frac{n+1}{2}} \tag{7}$$

and

$$L_1(\mathbf{Py}, \mathbf{PZ}) = \beta^k C|\mathbf{P}[\tilde{\mathbf{y}}, \mathbf{Z}]\mathbf{P}_{n+1}^{\perp}[\tilde{\mathbf{y}}, \mathbf{Z}]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}|^{-\frac{n+1}{2}} \tag{8}$$

where C is a non-zero constant, $\tilde{\mathbf{y}} = \beta^{-1}(\mathbf{y} - \alpha\mathbf{t})$ and $\mathbf{P}_{n+1}^{\perp} = \mathbf{I}_{n+1} - \mathbf{1}_{n+1}\mathbf{1}_{n+1}^{\mathrm{T}}/(n+1)$. Denote

$$\mathbf{A} = \mathbf{P}[\tilde{\mathbf{y}}, \mathbf{Z}]\mathbf{P}_{n+1}^{\perp}[\tilde{\mathbf{y}}, \mathbf{Z}]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}} \tag{9}$$

$$\mathbf{B} = \mathbf{P}[\mathbf{y}, \mathbf{Z}]\mathbf{P}_{n+1}^{\perp}[\mathbf{y}, \mathbf{Z}]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}. \tag{10}$$

Then the GLR can be given by

$$R(\tilde{\mathbf{y}}, \mathbf{Z}) = \frac{1}{\beta^k}\frac{|\mathbf{B}|^{\frac{n+1}{2}}}{|\mathbf{A}|^{\frac{n+1}{2}}}. \tag{11}$$

Remark 2: Similar to Remark 1, conditioning on a given $\mathbf{P}$, $(\mathbf{P}\tilde{\mathbf{y}}, \mathbf{Pz}_1, \mathbf{Pz}_2, \ldots, \mathbf{Pz}_n)$ is a simple random sample with size $n+1$ from some k-dimensional population, then, under a weak condition [53], its sample covariance matrix $(1/n+1)\mathbf{A}$ is non-singular with probability 1 as $k \leq \min\{n, d\}$. Similarly, $\mathbf{B}$ is also non-singular with probability 1.

Lemma 1: The GLR (11) can be rewritten in the following form:

$$R(\tilde{\mathbf{y}}, \mathbf{Z}) = \frac{1}{\beta^k}\left(1 + nT^2(\tilde{\mathbf{y}}, \mathbf{Z})\right)^{\frac{n+1}{2}} \tag{12}$$

where

$$T^2(\tilde{\mathbf{y}}, \mathbf{Z}) = \left(\mathbf{P}(\mathbf{y} - \tilde{\mathbf{y}})\right)^{\mathrm{T}}\left(\mathbf{P}\widehat{\Sigma}\mathbf{P}^{\mathrm{T}}\right)^{-1}\left(\mathbf{P}(\mathbf{y} - \tilde{\mathbf{y}})\right) \tag{13}$$

$$\widehat{\Sigma} = (n+1)[\tilde{\mathbf{y}}, \mathbf{Z}]\mathbf{P}_{n+1}^{\perp}[\tilde{\mathbf{y}}, \mathbf{Z}]^{\mathrm{T}}. \tag{14}$$

Proof: See Appendix B. □

It is worth noticing that the form of $T^2(\tilde{\mathbf{y}}, \mathbf{Z})$ is consistent with Hotelling's test statistic for the projection problem in [50], which seems to imply that we can choose a projection dimension $k = \min\{d, \lfloor(n/2)\rfloor\}$, where $\lfloor\cdot\rfloor$ represents the standard floor function. Under some assumptions, $k = \lfloor(n/2)\rfloor$ may attain the asymptotically optimal detection efficacy. $T^2(\tilde{\mathbf{y}}, \mathbf{Z})$ given by (13) can be viewed as the weighted Mahalanobis distance between $\mathbf{y} - \tilde{\mathbf{y}}$ and 0. Under the same assumption of $\mathbf{y}$ and $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n$ being i.i.d., it can be proved that $\mathbf{y} - \tilde{\mathbf{y}}$ is independent of $\widehat{\Sigma}$ by orthogonal transformation. Moreover, we observe that the GLR $R(\tilde{\mathbf{y}}, \mathbf{Z})$ is independent of the degrees of freedom ν. When ν tends to infinity, the matrix-variate t-distribution converges to matrix-variate normal distribution. Consequently, $T^2(\tilde{\mathbf{y}}, \mathbf{Z})$ has similar asymptotic properties to the test statistic under the projection problem in [50].

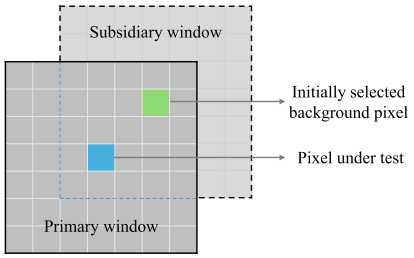

Fig. 1. Sliding dual-window used in SSDW strategy.

C. SSDW Strategy

In hyperspectral image processing, local detection is favored due to the intricate nature of background variability. Typically, sample characterizing the background is gathered around the pixel under test using a sliding dual-window technique, which employs a guard window and an outer window to delineate the area of interest. The statistical properties of the background, such as covariance, are estimated from the pixels in the annular space between these windows [56]. However, this direct estimation from the pixels within the dual-window boundaries may yield poor results (see [57], [58]). This is attributed to the proximity of these pixels to the potential target, causing them to act as statistical outliers and thus distort the background information. Specially, a small dual window is inadequate for targets that are closely spaced.

We introduce a novel SSDW strategy that utilizes a sliding dual-window mechanism and a similarity metric to isolate pixels for accurately representing the background. As shown in Fig. 1, the primary window is centered on the pixel under test with a pre-specified size (depending on practical applications), and then a subsidiary window is adaptively determined whose center has smallest similarity to the target signature than all other pixels in the primary window.

The metric of similarity between pixels is designed to jointly utilize the normalized Euclidean distance [59] and the Fisher–Rao distance [60] between two spectrum vectors. The former captures the absolute reflection intensity, while the latter represents geodesic distance within high-dimensional spectral space and measures the dissimilarity of the spectra by treating the normalized spectra as probability vectors. Specifically, for given m spectral vectors $\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{id}]^{\mathrm{T}} (i = 1, \ldots, m)$, the similarity between $\mathbf{x}_i$ and the other spectral vector $\mathbf{y} = [y_1, y_2, \ldots, y_d]^{\mathrm{T}}$ is defined as

$$\mathrm{Sim}(\mathbf{x}_i, \mathbf{y}) = \exp\left(-\frac{D_{\mathrm{Eud}}(\mathbf{x}_i, \mathbf{y}) \times D_{\mathrm{Fis}}(\mathbf{x}_i, \mathbf{y})}{\sigma}\right) \tag{15}$$

where

$$D_{\mathrm{Eud}}(\mathbf{x}_i, \mathbf{y}) = \|\mathbf{x}_i - \mathbf{y}\|^2 \tag{16}$$

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$$D_{\mathrm{Fis}}(\mathbf{x}_i, \mathbf{y}) = 2\arccos\left(\sum_{j=1}^{d}\sqrt{\frac{x_{ij}}{\sum_{j=1}^{d} x_{ij}}}\sqrt{\frac{y_j}{\sum_{j=1}^{d} y_j}}\right) \tag{17}$$

$$\sigma = \frac{1}{m}\sum_{i=1}^{m} D_{\mathrm{Eud}}(\mathbf{x}_i, \mathbf{y}) D_{\mathrm{Fis}}(\mathbf{x}_i, \mathbf{y}). \tag{18}$$

Using the proposed sliding dual-window and the above similarity metric, we provide Algorithm 1 for selecting the background sample relative to the given pixel under test.

Algorithm 1 Selection of the Background Sample Using SSDW Strategy

Input: A pixel under test and a target signature $\mathbf{t} \in \mathbb{R}^d$

Output: The sample $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n] \in \mathbb{R}^{d\times n}$ to represent the background relative to the pixel under test

1: Choose a squared primary window centered on the pixel under test with a pre-specified size
2: Calculate the similarities between all pixels within the primary window and the target signature $\mathbf{t}$
3: Select an initial background pixel $\mathbf{z}_1$ attaining the minimum in the above similarities
4: Choose a squared subsidiary window with center $\mathbf{z}_1$ and appropriate size
5: Calculate the similarities between the pixel $\mathbf{z}_1$ and other pixels in the subsidiary window
6: Select the pixels corresponding to the first $n-1$ largest similarities from step 5, denoted as $\mathbf{z}_2, \mathbf{z}_3, \ldots, \mathbf{z}_n$

Note that Algorithm 1 significantly reduces the risk of misidentifying target pixels as background in complex and varied environments. Moreover, it leverages the given target information to select a sample with minimal similarity to the target, thereby refining the depiction of the background and enhancing the performance of target detection.

D. RP-Based Sub-Pixel Target Detection

Fig. 2 is a flowchart of the proposed sub-pixel target detection using RP and SSDW strategy. Algorithm 2 summarizes the complete computation procedure of GLRs for three observation models.

Algorithm 2 GLRs for RP-based Sub-pixel Target Detection

Input: The hyperspectral image data $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N] \in \mathbb{R}^{d\times N}$ with N pixels and target signature $\mathbf{t} \in \mathbb{R}^d$

Output: The GLR R_{PA}, R_{PR} or R_{PM} for one of three models

1: **for** $i = 1$: N **do**
2: Select a background sample $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n] \in \mathbb{R}^{d\times n}$ using Algorithm 1
3: Generate a random matrix $\mathbf{P} \in \mathbb{R}^{k\times d}$
4: Calculate the corresponding GLR via (4), (5) or (6)
5: **end**

Note that Algorithm 2 is a non-iterative algorithm since each session is done by a one-time computation.

E. Computational Complexity

We can find, from Algorithm 2, that the detection procedure for a pixel under test can be roughly divided into three steps. Assume the sizes of the primary and subsidiary windows are $m_1 \times m_1$ and $m_2 \times m_2$, respectively. In the first step, the similarity vector involves $(m_1^2 + m_2^2 - 2)$ distances of two d-dimensional vectors, each of which has the same operation complexity denoted as $\mathcal{O}(c_1)$. Then the computational complexity of the session is $\mathcal{O}(c_1(m_1^2 + m_2^2 - 2))$. In the second and third steps, which of computational complexity is denoted as $\mathcal{O}(c_2)$, the main operations include the inversion of a k-dimensional symmetric and positive matrix, multiplication of matrices and vectors, and optimization of two unknown scalar parameters α and β in (5) and (6). From the discussion in the last part of Section II-A, we know that the optimal α (or β) is a solution of a quadratic equation. Therefore, for an image consisting of N pixels, the computational complexity of the proposed detectors can be approximated as $\mathcal{O}(N(c_1(m_1^2 + m_2^2 - 2) + c_2))$. While $m_1 \ll N$, $m_2 \ll N$, c_1, and c_2 are small, the proposed detectors possess a linear complexity with respect to N. Such efficiency will be verified in the next experiments.

III. Experiments and Analyses

In this section, we will illustrate the performance of the proposed approach on some real hyperspectral image datasets and synthetic data. The raw hyperspectral image data undergo min-max normalization [61]. All experiments are implemented in Octave on a computer with AMD Ryzen 7 5800H 3.2 GHz processor.

A. Experiment Settings

To evaluate the performance of the proposed detectors (named RP-KELLY, RP-ACUTE, and RP-SPADE, derived from the GLRs (4)–(6), respectively), we consider seven existing GLRT-based detectors including the ACE [39], MF [28], FTMF [28], EC-FTMF [44], KELLY, ACUTE, and SPADE [43] for comparison purpose. Note that the background following Gaussian distribution is assumed in the ACE, MF, and FTMF. The EC-FTMF is an extension of the FTMF for matrix-variate t-distribution background, where the degrees of freedom ν is set to 3 in the next. And the KELLY, ACUTE, and SPADE also assume that the background obeys a matrix-variate t-distribution. Additionally, we also compare several state-of-the-art detectors: CEM [33], HCEM [34], ECEM [35], KCEM [36], KTCIMF [32], CTTD [62], and CGSAL [63], which do not make any assumption on the background distribution.

The detection performance is evaluated through the 3-D receiver operating characteristic (ROC) curve as a measure [64]. It portrays the relationship between the detection probability $\mathrm{P_D}$, the false alarm probability $\mathrm{P_F}$, and the threshold τ, leading to three 2-D ROC curves in ($\mathrm{P_D}$, $\mathrm{P_F}$), ($\mathrm{P_D}$, τ), and ($\mathrm{P_F}$, τ) planes. Moreover, we also use several intuitive metrics $\mathrm{AUC_{(D,F)}}$, $\mathrm{AUC_{(D,\tau)}}$, $\mathrm{AUC_{(F,\tau)}}$, $\mathrm{AUC_{TD}}$, $\mathrm{AUC_{BS}}$, $\mathrm{AUC_{TDBS}}$, $\mathrm{AUC_{SNPR}}$, and $\mathrm{AUC_{ODP}}$ introduced in [64] and [65] as evaluation criteria.

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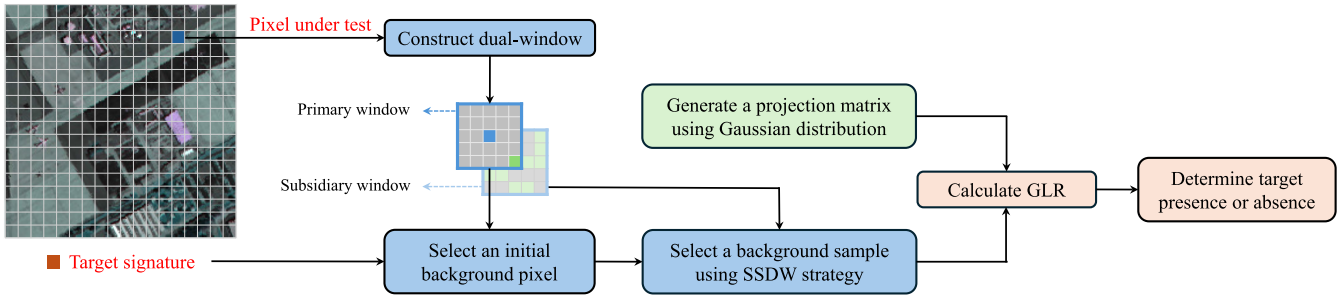

Fig. 2. Flowchart of RP-based sub-pixel target detection.

TABLE I

SIZES OF DUAL WINDOW AND THE CORRESPONDING SAMPLE SIZES FOR SAN DIEGO AIRPORT IMAGE

Primary Window	10×10	10×10	10×10	10×10	10×10
Subsidiary Window	20×20	22×22	24×24	26×26	28×28
Guard Window	5×5	5×5	5×5	5×5	5×5
Outer Window	15×15	17×17	19×19	21×21	23×23
n	200	264	336	416	504

TABLE II

SIZES OF DUAL WINDOW AND THE CORRESPONDING SAMPLE SIZES FOR HYMAP IMAGE

Primary Window	10×10	10×10	10×10	10×10	10×10
Subsidiary Window	18×18	20×20	22×22	24×24	26×26
Guard Window	5×5	5×5	5×5	5×5	5×5
Outer Window	13×13	15×15	17×17	19×19	21×21
n	144	200	264	336	416

TABLE III

SIZES OF DUAL WINDOW AND THE CORRESPONDING SAMPLE SIZES FOR SALINAS IMAGE

Primary Window	10×10	10×10	10×10	10×10	10×10
Subsidiary Window	22×22	24×24	26×26	28×28	30×30
Guard Window	5×5	5×5	5×5	5×5	5×5
Outer Window	17×17	19×19	21×21	23×23	25×25
n	264	336	416	504	600

Besides the proposed SSDW strategy, the traditional sliding dual-window method with a guard window and an outer window is considered for comparison purposes. Tables I–III provide experiment settings of the sizes of dual window and the corresponding sample sizes for San Diego airport, HyMap, and Salinas images, respectively. The approximate range of these window sizes usually depends on prior knowledge of the specific target size and the resolution of the image [54], [56], [58].

B. San Diego Airport Experiments

The San Diego airport public data is collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over San Diego, CA, USA, which can be accessed from the website at http://aviris.jpl.nasa.gov/. This hyperspectral image contains prior information regarding the target spectrum and offers a spatial resolution of 3.5 m, with a wavelength range of 370 to 2510 nm. Initially comprising 224 spectral bands, the image was refined to 189 bands after excluding those with low signal-to-noise ratios and other inferior-quality bands. The spatial size of the entire image is 400×400, as depicted in Fig. 3(a). Two sub-images, namely SDA-I and SDA-II, respectively, having sizes of 200×200 [see Fig. 3(b)] and 100×100 [see Fig. 3(c)], are selected as the datasets for target detection. Our task is to detect the aircraft in the airport. The ground truths of SDA-I and SDA-II are given for evaluating the detection performance, which is shown in Fig. 3(d) and (e), respectively.

1) SDA-I Experiment: For the SDA-I dataset, the AUCs for all detectors across different sample sizes are reported in Table IV (hereinafter, the optimal value on each row is bolded). The 3-D ROC curve under sample size $n = 264$, along with the corresponding 2-D ROC curves, are demonstrated in Fig. 4. It can be observed that the detectors based on RP, particularly RP-ACUTE and RP-SPADE, exhibit superior performance. All the other detectors compared show unsatisfactory performance for a small sample size, but their efficiency improves with increasing sample size. This trend can be attributed to two factors: 1) the background of the area to be tested is relatively complex and the background spectrum is similar to the target spectrum, which leads to difficulties in discovering the location of the targets and 2) when the sample size is close to the spectral dimension, the background covariance matrix estimate may be unstable or even singular, resulting in a degradation of the detection performance. Surprisingly, the FTMF does not perform as expected, whereas the EC-FTMF shows promising results. It suggests that the heavy-tailed t-distribution more accurately portrays the background than the Gaussian distribution. As indicated in Table IV, RP-ACUTE and RP-SPADE show better detection performance in general, in contrast to RP-KELLY, which is slightly worse. This exception may be due to a poor estimate of target abundance, because the additive model is strongly limited by the abundance of the background spectrum being 1 in finding the unknown abundance of the target spectrum. This experiment clearly demonstrates that the detectors operating under t-distribution background outperform those using a Gaussian distribution background.

2) SDA-II Experiment: The above SDA-I experiment assumes that the target spectrum directly comes from ground truth. However, it may be hard to ensure in practice. Notably, some detectors such as ECEM and KTCIMF are sensitive to the pre-given ground truth of the target. Therefore, we take an incomplete ground truth as a target to explore its effect

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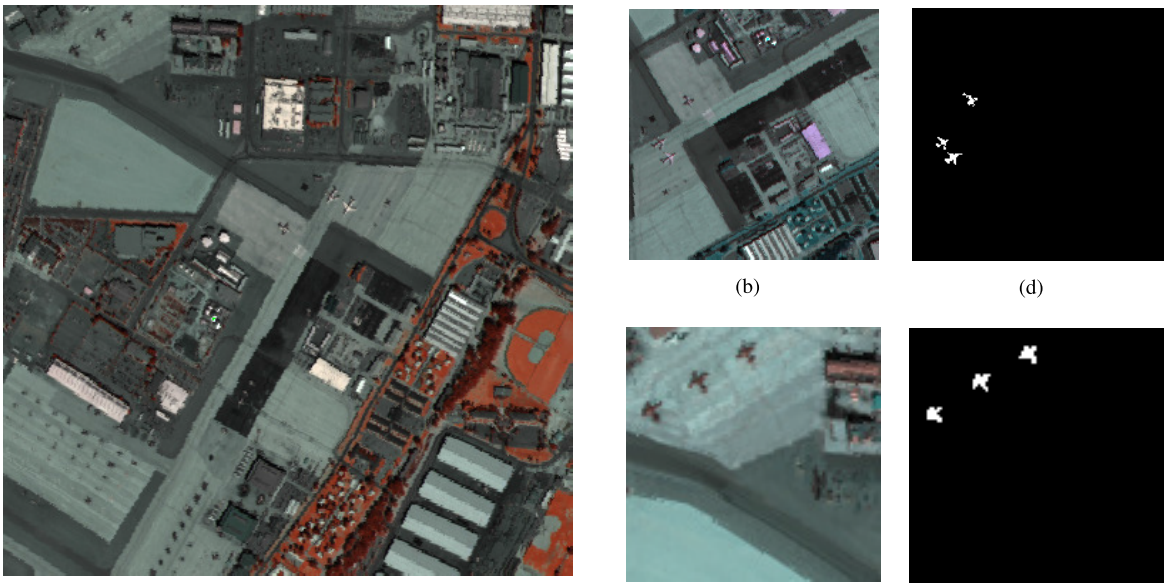
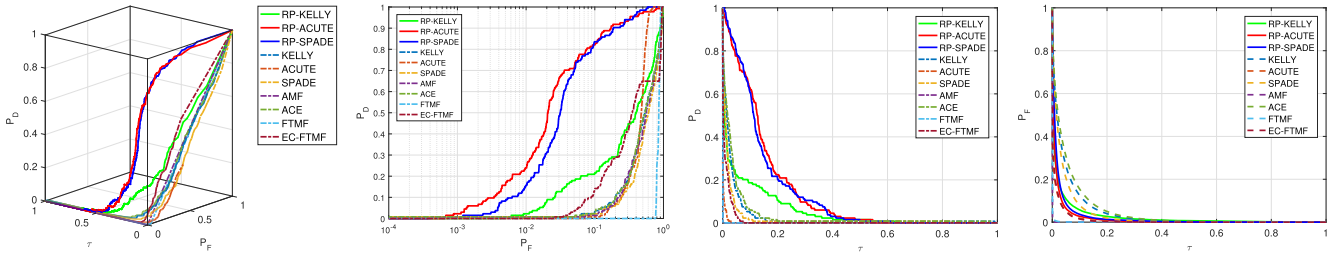


(a) (b) (c) (d) (e)

Fig. 3. San Diego airport dataset. (a) Complete false-color image, (b) false-color image of SDA-I, (c) false-color image of SDA-II, (d) ground-truth map of SDA-I, and (e) ground-truth map of SDA-II.



(a) (b) (c) (d)

Fig. 4. ROC curves under sample size $n = 264$ on SDA-I image. (a) 3-D ROC curve, (b) 2-D ROC curve of (P_D, P_F), (c) 2-D ROC curve of (P_D, τ), and (d) 2-D ROC curve of (P_F, τ).

TABLE IV
AUCs OF DIFFERENT DETECTORS ON SDA-I IMAGE

n	AMF	ACE	FTMF	EC-FTMF	KELLY	ACUTE	SPADE	RP-KELLY	RP-ACUTE	RP-SPADE
200	0.1426	0.3408	0.1015	0.2907	0.3392	0.2727	0.2423	0.5607	**0.9627**	0.9597
264	0.4303	0.4432	0.1536	0.5278	0.4418	0.5581	0.3954	0.5514	**0.9295**	0.9283
336	0.5601	0.5320	0.1559	0.5902	0.5426	0.4894	0.5154	0.5823	0.8525	**0.8576**
416	0.5948	0.5493	0.1555	0.6076	0.5712	0.4120	0.5420	0.5913	0.7506	**0.7996**
504	0.6188	0.5681	0.1529	0.6439	0.5973	0.3440	0.5647	0.6172	0.6515	**0.7369**

on the detection performance. Specifically, two scenarios are considered, i.e., target spectral signatures are estimated from randomly selected pixels of 10% and 30% out of 85 pixels labeled as airplanes, respectively. The sample size is set to $n = 144$ in our proposed methods.

Figs. 5 and 6 illustrate the detection maps of each detector in two scenarios. Visually, the HCEM, ECEM, KCEM, and TCIMF are strongly affected by the number of randomly selected pixels and show poor detection performance. In contrast, our proposed methods, especially RP-ACUETE and RP-SPADE, are only slightly affected by such numbers and can highlight targets of interest. However, they are inferior to other detectors in terms of background suppression. Remarkably, the CGSAL performs relatively well in terms of both detection performance and efficiency. Fig. 7 shows the 3-D ROC curves and the corresponding 2-D ROC curves generated from the detection results of Figs. 5 and 6, respectively, while Tables V and VI demonstrate the different AUCs for two scenarios. Overall, the RP-SPADE exhibits excellent detection performance. Interestingly, the detection performance of the compared methods improves as the number of randomly selected pixels increases, while the same phenomenon does not seem to occur with our proposed method. In addition, the KCEM and KTCIMF are computationally expensive, whereas the CEM and CTTD are very efficient detectors.

To further portray the degree of separation between target and background, target-background separation map is commonly adopted [5], [66], [67]. Figs. 8(a) and (b), respectively,

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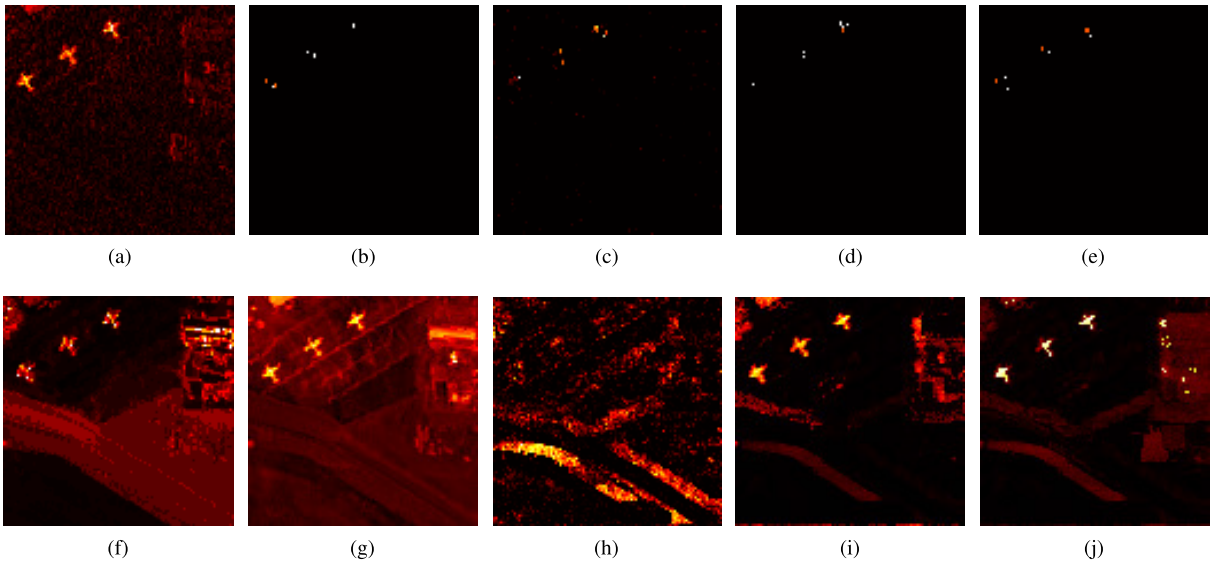

Fig. 5. Detection maps on SDA-II image with 10% randomly selected pixels labeled as airplanes to calculate target spectrum. (a) CEM, (b) HCEM, (c) ECEM, (d) KCEM, (e) KTCIMF, (f) CTTD, (g) CGSAL, (h) RP-KELLY, (i) RP-ACUTE, and (j) RP-SPADE.

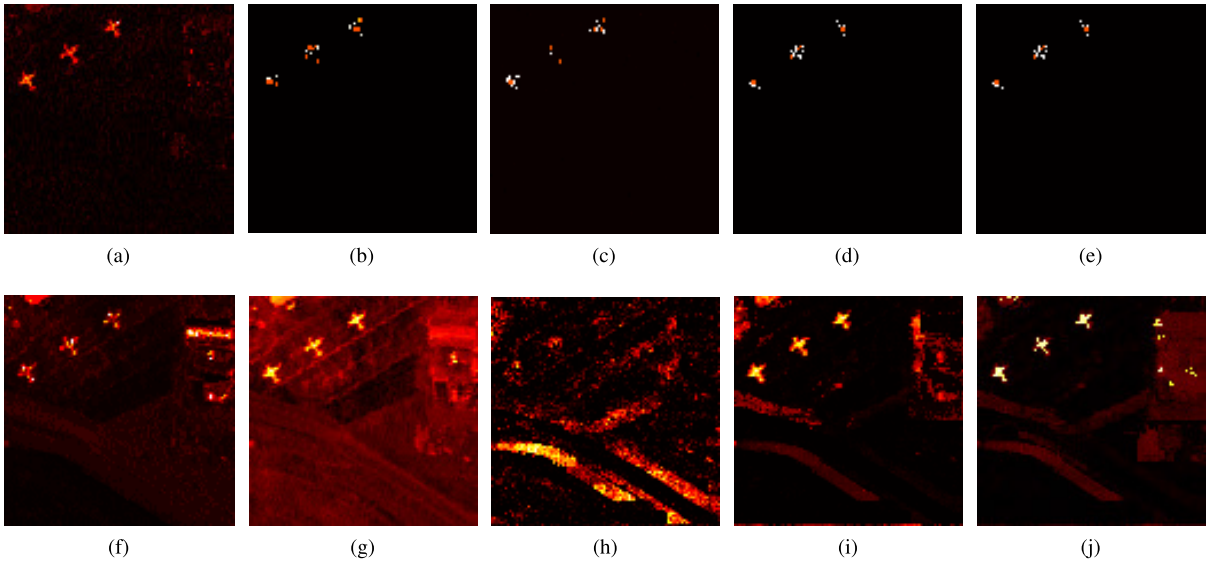

Fig. 6. Detection maps on SDA-II image with 30% randomly selected pixels labeled as airplanes to calculate target spectrum. (a) CEM, (b) HCEM, (c) ECEM, (d) KCEM, (e) KTCIMF, (f) CTTD, (g) CGSAL, (h) RP-KELLY, (i) RP-ACUTE, and (j) RP-SPADE.

TABLE V

AUCs WITH 10% RANDOMLY SELECTED PIXELS LABELED AS AIRPLANES TO CALCULATE TARGET SPECTRUM ON SDA-II IMAGE

Performance	CEM	HCEM	ECEM	KCEM	KTCIMF	CTTD	CGSAL	RP-KELLY	RP-ACUTE	RP-SPADE
$AUC_{(D,F)}$	0.9423	0.5588	0.6898	0.5529	0.5706	0.9312	0.9724	0.6884	**0.9878**	0.9768
$AUC_{(D,\tau)}$	0.3903	0.0987	0.1119	0.0982	0.0984	0.3903	0.5381	0.1178	0.5296	**0.5495**
$AUC_{(F,\tau)}$	0.0881	**0.0050**	0.0055	**0.0050**	**0.0050**	0.0881	0.1263	0.0648	0.0284	0.0328
AUC_{TD}	1.3326	0.6575	0.8017	0.6511	0.6690	1.3215	1.5105	0.8062	1.5174	**1.5263**
AUC_{BS}	0.8542	0.5538	0.6843	0.5479	0.5656	0.8431	0.8461	0.6236	**0.9594**	0.9440
AUC_{TDBS}	0.3022	0.0937	0.1064	0.0932	0.0934	0.3022	0.4118	0.0530	0.5012	**0.5167**
AUC_{SNPR}	4.4302	19.7400	**20.3455**	19.6400	19.6800	4.4302	4.2605	1.8179	18.6479	16.7530
AUC_{ODP}	1.2445	0.6525	0.7962	0.6461	0.6640	1.2334	1.3842	0.7414	1.4890	**1.4935**
Time(s)	**0.1037**	0.2269	11.5426	568.8629	573.1221	0.1880	3.9729	43.8815	44.8991	45.1828

show the separation maps of the different target detectors for the two scenarios. The red and blue boxes represent the normalized detection statistics for target and background pixels, respectively. The upper and lower boundaries of the boxes represent the first and third quartiles of the normalized detection statistics, respectively. Additionally, the upper and lower horizontal lines, respectively, represent the maximum and minimum. It can be seen that the RP-SPADE and

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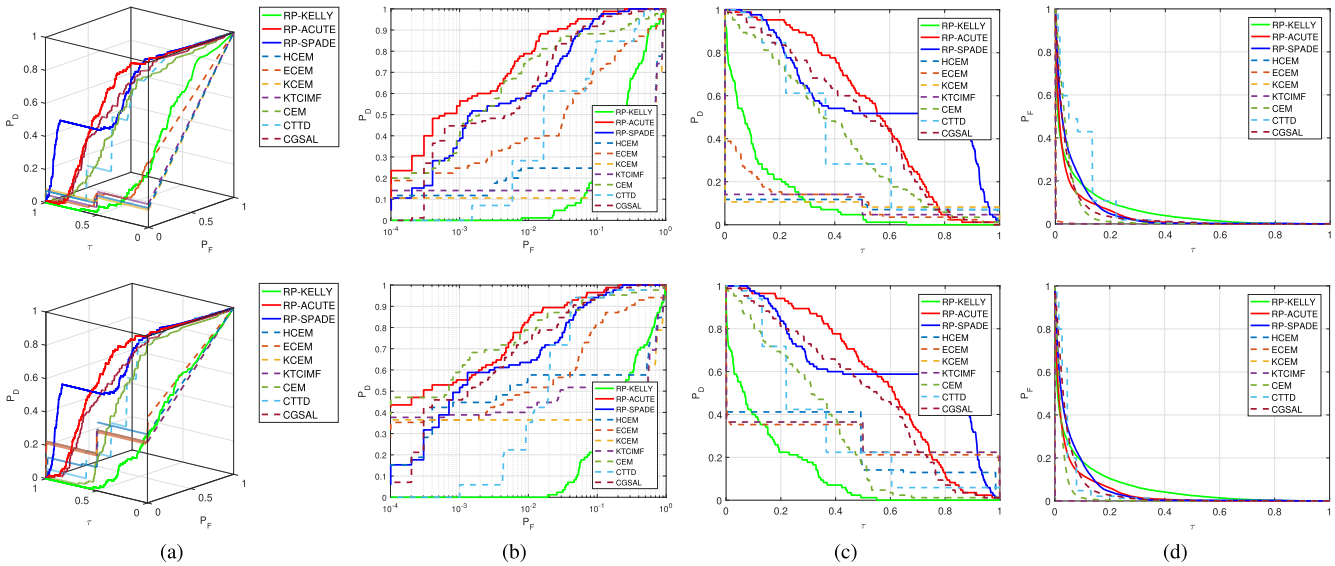

Fig. 7. ROC curves with randomly selected pixels labeled as airplanes to calculate target spectrum on SDA-II image. First row: 10% of sampling rate; second row: 30% of sampling rate. (a) 3-D ROC curve, (b) 2-D ROC curve of (P_D, P_F), (c) 2-D ROC curve of (P_D, τ), and (d) 2-D ROC curve of (P_F, τ).

TABLE VI

AUCs WITH 30% RANDOMLY SELECTED PIXELS LABELED AS AIRPLANES TO CALCULATE TARGET SPECTRUM ON SDA-II IMAGE

Performance	CEM	HCEM	ECEM	KCEM	KTCIMF	CTTD	CGSAL	RP-KELLY	RP-ACUTE	RP-SPADE
$AUC_{(D,F)}$	0.9629	0.7057	0.6765	0.6824	0.6824	0.9709	0.9814	0.6585	**0.9890**	0.9765
$AUC_{(D,\tau)}$	0.3120	0.2721	0.2918	0.2955	0.2958	0.3286	0.5662	0.1255	0.5667	**0.5892**
$AUC_{(F,\tau)}$	0.0236	0.0052	0.0151	**0.0050**	**0.0050**	0.0486	0.1377	0.0667	0.0286	0.0317
AUC_{TD}	1.2749	0.9778	0.9683	0.9779	0.9782	1.2995	1.5476	0.7840	1.5557	**1.5657**
AUC_{BS}	0.9393	0.7005	0.6614	0.6774	0.6774	0.9223	0.8437	0.5918	**0.9604**	0.9448
AUC_{TDBS}	0.2884	0.2669	0.2767	0.2905	0.2908	0.2800	0.4285	0.0588	0.5381	**0.5575**
AUC_{SNPR}	13.2203	52.3269	19.3245	59.1000	**59.1600**	6.7613	4.1118	1.8816	19.8147	18.5868
AUC_{ODP}	1.2513	0.9726	0.9532	0.9729	0.9732	1.2509	1.4099	0.7173	1.5271	**1.5340**
Time(s)	**0.1004**	18.6881	13.7947	569.2337	568.4777	0.1712	4.0422	43.2485	44.0293	45.4562

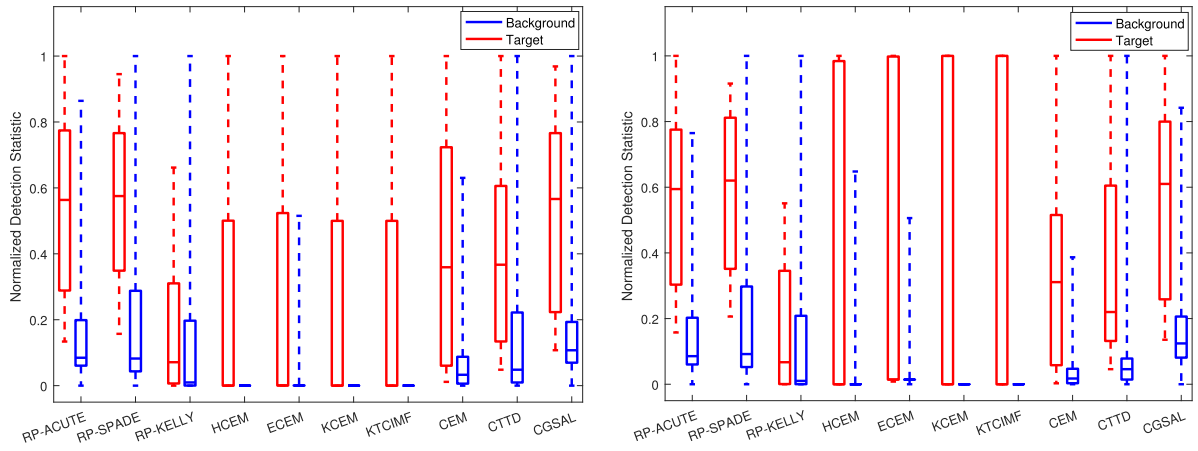

Fig. 8. Separation map with randomly selected pixels labeled as airplanes to calculate target spectra on SDA-II image. (a) 10% of sampling rate and (b) 30% of sampling rate.

RP-ACUTE have good target and background separation in both cases, and their performance is almost independent of the number of randomly selected pixels. On the contrary, the HCEM, ECEM, KCEM, and KTCIMF are more affected by such numbers, although they all have good suppression of background. In summary, the CGSAL, RP-SPADE, and RP-ACUTE perform a better ability to separate target from background.

C. HyMap Image Experiment

In this section, we consider the dataset collected by the Rochester Institute of Technology (RIT) employing the HyMap sensor that can be accessed from the website at http://dirsapps.cis.rit.edu/blindtest/. This is an open-source dataset gathered during a field study in the vicinity of the small town of Cooke, MT, USA, in July 2006 [68]. It includes both a standard self-test dataset and a dataset without target

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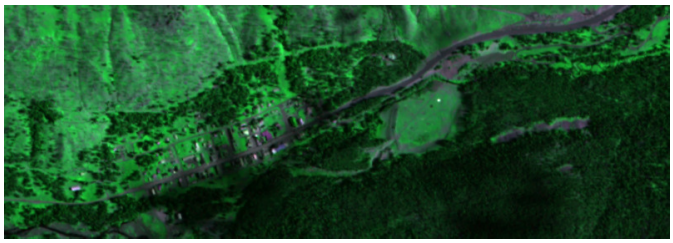

Fig. 9. False-color image of HyMap.

TABLE VII
TARGETS WITH GROUND TRUTH IN THE SELF-TEST DATASET

Target	Size(m)	Description
V1	4×2	Light green Chevrolet Trailblazer
V2	3×1.7	White Toyota T100
V3	4.5×1.6	Red Subaru GL Wagon
F1	3×3	Red cotton fabric panel
F2	3×3	Yellow nylon fabric panel
F3	1×1 and 2×2	Blue cotton fabric panel
F4	1×1 and 2×2	Red nylon fabric panel

positions. The self-test dataset, which is utilized in the next experiment, derives from an image with approximately 3×3 m spatial resolution, encompassing 280×800 pixels, each of which features 126 bands (refer to Fig. 9). Table VII details the target descriptions within this dataset. These targets are typically considered sub-pixels because it is obvious that their sizes are comparable to the spatial resolution [69]. For the seven different targets, our experiments focus on three kinds of specific targets: V3 (vehicle), F1, and F2 (cotton panel).

The AUCs and 3-D ROC curves of all detectors across the various targets are shown in Table VIII and Fig. 10, respectively. Notably, the detectors based on RP, prefixed with RP, generally enhance detection performance compared to others like KELLY, ACUTE, and SPADE. The RP-ACUTE, in particular, shows remarkable effectiveness in detecting all three targets. Furthermore, detectors like FTMF, EC-FTMF, and ACUTE, which are based on replacement models, demonstrate substantial efficacy. It is attributed that the replacement model-based detectors can cross-select the best fill factor between the background and target owing to the reduction in the probability of false alarms. For the case of a small sample size, the RP-KELLY, RP-ACUTE, and RP-SPADE exhibit superior performance in detecting V3 and F1. As the sample size grows, the performance gap among these detectors narrows. It indicates that their effectiveness is more pronounced when the sample size is small relative to the spectral dimension. Besides, an overall observation from the experiments reveals the lower AUCs for detecting V3 compared to F1 and F2, which suggests that the spectrum of V3 might more closely resemble the background spectra.

D. Salinas Image Experiments

We take into account synthetic data generated from a real hyperspectral image [70]. The raw dataset is collected using 224-band AVIRIS sensor over Salinas Valley, CA, USA, and can be accessed from the GIC website at http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes. It encompasses an area represented by 512×217 pixels, each with a spatial resolution of 3.7 m, as illustrated in Fig. 11(a). This region contains vegetables, bare soil, and vineyard fields. As in [70], from the original dataset, 20 water-absorbed bands (108–112, 154–167, and 224) are excluded, leaving 204 bands available for analysis.

In the following experiments, a specific 120×120 sub-image of interest, depicted in Fig. 11(b), in the Salinas hyperspectral image is chosen. In this region, some buildings that markedly contrast with their surroundings are selected as detection targets. Then, their spectra are implanted into the Salinas image by using the popular target implantation method [71]. To realistically simulate anomalous targets in real images, we conduct simulations for two scenarios of linearly and nonlinearly mixing the target and background.

1) Linear Simulation Experiment: In this experiment, 25 pixels within the cropped region are randomly selected as anomalous targets. Subsequently, the LMM is applied to synthesize the spectra of sub-pixel targets as [70]

$$\mathbf{y} = \gamma \times \mathbf{t} + (1 - \gamma) \times \mathbf{b} \tag{19}$$

where $\mathbf{y} \in \mathbb{R}^d$ is the spectrum of the synthesized sub-pixel target, γ is the specified abundance ranging from 0.04 to 1 with a step size of 0.04, and $\mathbf{t} \in \mathbb{R}^d$ and $\mathbf{b} \in \mathbb{R}^d$ are the target and background spectra, respectively. The ground truth of these anomalous targets is presented in Fig. 11(c).

From the AUCs reported in Table IX, the detectors based on RP (prefixed with RP) consistently outperform the others across various sample sizes. Meanwhile, the detection performance of these detectors also demonstrates the following phenomenon: it improves as the sample size increases. However, it is noticeable that when the spectral dimension exceeds the sample size (as in cases $n = 144$ and $n = 200$ in Table IX), the sample covariance matrix becomes singular, which renders the other detectors ineffective. On the contrary, our proposed detectors overcome this limitation by testing projected samples in low-dimensional space, thus remaining functional and exhibiting substantial performance even under these situations.

In addition, as illustrated in Fig. 12, the 2-D ROC curves of (P_D, P_F) for RP-KELLY, RP-ACUTE, and RP-SPADE are almost overlapping and show the best performance. It is obvious that these detectors maintain a high P_D at very low P_F. Coincidentally, the EC-FTMF, ACUTE, and SPADE, are also derived under t-distributed background and demonstrate strong performance. This can be attributed to the assumption that the background follows t-distribution, which more accurately reflects the background characteristics than normal distribution. Furthermore, Fig. 13 shows the detection maps of various methods. As we can see, our proposed detectors can detect almost all targets and perform well in background suppression, especially RP-ACUTE and RP-SPADE.

Table X reports the elapsed computation times of the detection algorithms. In general, our proposed detectors, prefixed with RP, excel in computational efficiency. Specifically, the time they consume increases linearly with sample size. In contrast, the amount of time required by the other detectors barely varies with the sample size and is greater than that of

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TABLE VIII

AUCs OF DIFFERENT DETECTORS FOR VARIOUS TARGETS ON HYMAP IMAGE

Target	n	AMF	ACE	FTMF	EC-FTMF	KELLY	ACUTE	SPADE	RP-KELLY	RP-ACUTE	RP-SPADE
V3	144	0.4030	0.4467	0.3723	0.3909	0.4466	0.4393	0.5331	0.8627	**0.9091**	0.7845
	200	0.6460	0.7504	0.6369	0.7029	0.7486	0.7083	0.7010	0.7558	**0.8631**	0.7182
	264	0.5519	0.6420	0.6397	0.6892	0.6355	0.7154	0.6026	0.9068	**0.9455**	0.8789
	336	0.5997	0.6834	0.7528	0.7931	0.6738	0.8188	0.6741	0.9403	**0.9670**	0.9255
	416	0.7243	0.7753	0.8435	0.8724	0.7718	0.8688	0.7673	0.8624	**0.9275**	0.8441
F1	144	0.6404	0.3252	0.3596	0.3758	0.3261	0.4188	0.2624	0.5685	**0.6567**	0.5813
	200	0.7344	0.4871	0.7157	0.6863	0.4947	0.6798	0.4224	0.5932	**0.7684**	0.4985
	264	0.9036	0.6948	0.9210	0.8198	0.7201	0.8128	0.6474	0.9235	**0.9539**	0.8885
	336	0.9608	0.8752	0.9770	0.9291	0.9052	0.9337	0.8600	0.9696	**0.9803**	0.9618
	416	0.9672	0.8859	**0.9768**	0.9295	0.9263	0.9414	0.8772	0.9566	0.9724	0.9502
F2	144	0.8438	0.7373	0.8819	0.8366	0.7380	0.8573	0.6800	0.9028	**0.9524**	0.8628
	200	0.9488	0.8139	0.9196	0.8966	0.8202	0.8853	0.7771	0.9706	**0.9857**	0.9573
	264	0.9830	0.9382	0.9882	0.9666	0.9468	0.9634	0.9293	0.9981	**0.9995**	0.9976
	336	0.9814	0.9407	0.9840	0.9669	0.9531	0.9664	0.9368	0.9933	**0.9968**	0.9924
	416	0.9819	0.9435	0.9842	0.9668	0.9610	0.9683	0.9410	0.9851	**0.9889**	0.9834

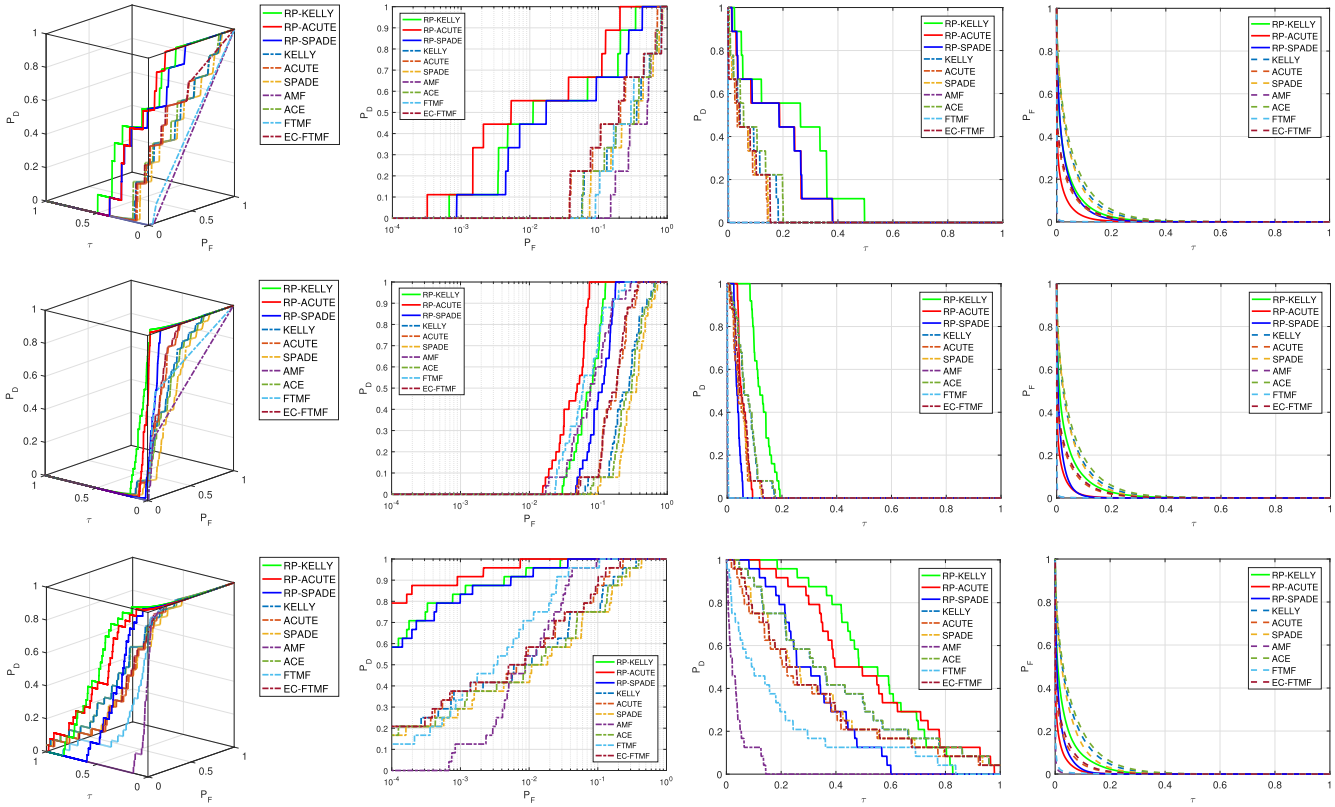

Fig. 10. ROC curves for detecting various targets under sample size $n = 264$ on HyMap image. First row: V3; second row: F1; third row: F2. (a) 3-D ROC curve, (b) 2-D ROC curve of (P_D, P_F), (c) 2-D ROC curve of (P_D, τ), and (d) 2-D ROC curve of (P_F, τ).

TABLE IX

AUCs OF DIFFERENT DETECTORS ON SALINAS IMAGE WITH LINEAR MIXING

n	AMF	ACE	FTMF	EC-FTMF	KELLY	ACUTE	SPADE	RP-KELLY	RP-ACUTE	RP-SPADE
144	–	–	–	–	–	–	–	**0.9999**	**0.9999**	**0.9999**
200	–	–	–	–	–	–	–	**0.9999**	**0.9999**	**0.9999**
264	0.7378	0.7928	0.8635	0.9238	0.7897	0.8546	0.9230	**0.9999**	**0.9999**	**0.9999**
336	0.8477	0.8878	0.9600	0.9898	0.8717	0.9899	0.9683	**0.9999**	**0.9999**	**0.9999**
416	0.9681	0.9793	0.9974	0.9989	0.9748	0.9986	0.9897	**0.9999**	**0.9999**	**0.9999**
504	0.9968	0.9976	0.9996	0.9998	0.9974	0.9998	0.9987	**0.9999**	**0.9999**	**0.9999**
600	0.9985	0.9994	**0.9999**	**0.9999**	0.9990	**0.9999**	0.9998	**0.9999**	**0.9999**	**0.9999**

"–" means that the detector can not execute properly.

the proposed detectors. This enhanced efficiency is attributed to two important reasons. One is that the determination of the projection dimension does not require any extra training; the other is that the RP-based detectors project the sample

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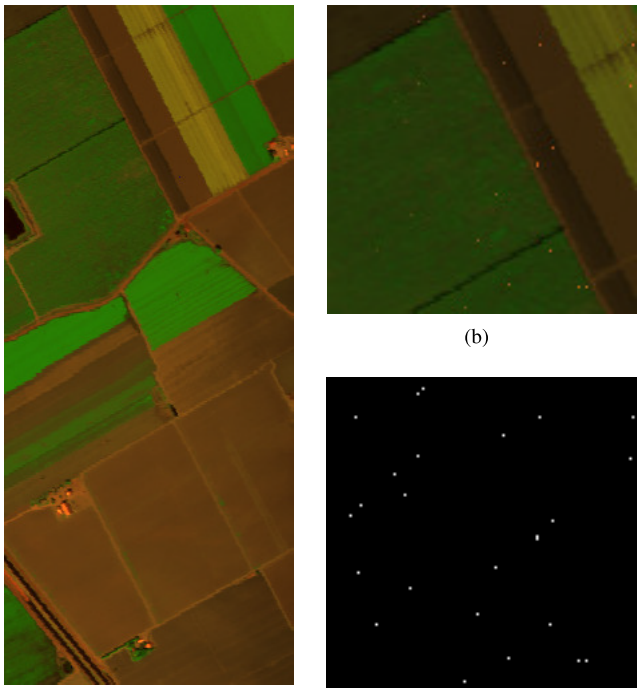
TABLE X
ELAPSED TIME ON SALINAS SYNTHETIC IMAGE WITH LINEAR MIXING IN SECOND

n	AMF	ACE	FTMF	EC-FTMF	KELLY	ACUTE	SPADE	RP-KELLY	RP-ACUTE	RP-SPADE
144	279.5302	281.0818	280.4404	280.5270	281.9520	281.6296	285.3229	**62.5680**	63.3483	64.4834
200	286.3157	286.8950	286.2702	287.1242	288.0502	287.1460	289.2827	85.8037	**85.0407**	87.4190
264	278.5474	278.7722	278.0344	279.0843	280.3217	278.8768	280.7201	**111.4853**	111.8481	114.3284
336	288.5302	289.5548	289.4096	288.4435	290.6638	289.4625	290.0538	**143.6185**	145.2670	149.8847
416	295.1825	295.0483	294.4715	295.5488	295.3049	295.6396	296.9870	**184.4943**	184.7666	191.4192



(a) (b) (c)

Fig. 11. Salinas image dataset. (a) Complete false-color image, (b) false-color image of the synthetic scene with linear mixing, and (c) ground-truth map of the synthetic scene with linear mixing.

into a lower-dimensional space, thereby consuming less time in executing the operations such as eigendecomposition and matrix inversion for the covariance matrix.

2) Nonlinear Simulation Experiment: In many real scenarios, the mixing of target and background is more likely to be nonlinear due to multiple scattering effects between photons. Therefore, similar to the classical bilinear mixing model [72], we simulate this nonlinear mixing by additionally incorporating an interaction part between the target and background in the linear model. Specifically, the nonlinear mixing model is

$$\mathbf{y} = \lambda \times \mathbf{t} + (1-\lambda) \times \mathbf{b} + \eta \times \lambda \times (1-\lambda) \times \mathbf{t} \odot \mathbf{b} \quad (20)$$

where $\lambda > 0$ is the mixing coefficient, $\eta > 0$ denotes the weighting coefficient combining the target and background, and the symbol $\odot$ represents the Hadamard product of two vectors.

In this experiment, 15 anomalous targets with different intensities and shapes are implanted into the synthetic image. These targets are located in three rows and five columns, and different rows have different shapes and sizes, while the same column has the same intensity. Specifically, the sizes of the targets from the first row to the third row are 1, 2, and 4, respectively. The mixing coefficient λ in different columns is 0.05, 0.1, 0.2, 0.4, and 0.6, respectively, and the corresponding weighting coefficient η between the target and background is set to 10^{-4}, 5×10^{-5}, 10^{-5}, 5×10^{-6}, and 10^{-6}, respectively. The false-color map of the synthetic scene and the corresponding ground truth are, respectively, shown in Fig. 14(a) and (b). Our proposed methods set the sample size to $n = 200$.

Fig. 15 demonstrates the 3-D ROC curve and the corresponding three 2-D ROC curves for the synthesis scene on Salinas image with nonlinear mixing, and Table XI presents the different AUC metrics for each detector and the calculation time. As can be seen from the detection results, the overall performance of the RP-KELLY is reasonably good. It is noticed that the HCEM has the strongest background suppression, while the RP-KELLY has the best detection performance. In addition, the CEM is still the most efficient algorithm, while kernel-based methods are extremely expensive in terms of computational complexity, due to the fact that they require the kernel transformation of the correlation matrix.

E. Evaluation of RP Method

In this section, we evaluate the effectiveness of the RP method in the sense of AUC. The sample for extracting background statistical information is selected by the SSDW strategy.

For the SDA-I image, the adoption of the RP markedly improves the detection performance for the case of $n/2 < d$, as illustrated in Table XII. However, when $n/2 > d$, the impact of this method on enhancing performance is not as pronounced. In the Salinas synthetic image with linear mixing, the detectors with RP consistently perform no worse than those without RP, irrespective of the sample size. Notably, the RP substantially enhances detection performance when the sample size approximates the spectral dimension. Not only that, but in terms of computational speed, RP reduces the dimensionality of the original sample for testing, thereby significantly reducing the computational burden. As demonstrated in Table X, the detectors employing RP outperform those that do not in terms of computational efficiency.

F. Comparison of Different Projection Dimensions

One of the basic parameters of the RP-based method is the projection dimension k. To verify its effect on the proposed detectors, different projection dimensions are selected for testing.

The targets selected for this investigation are the vehicle V3 and the cotton fabric panel F2 from the HyMap image, representing two distinct object types. The sizes of primary

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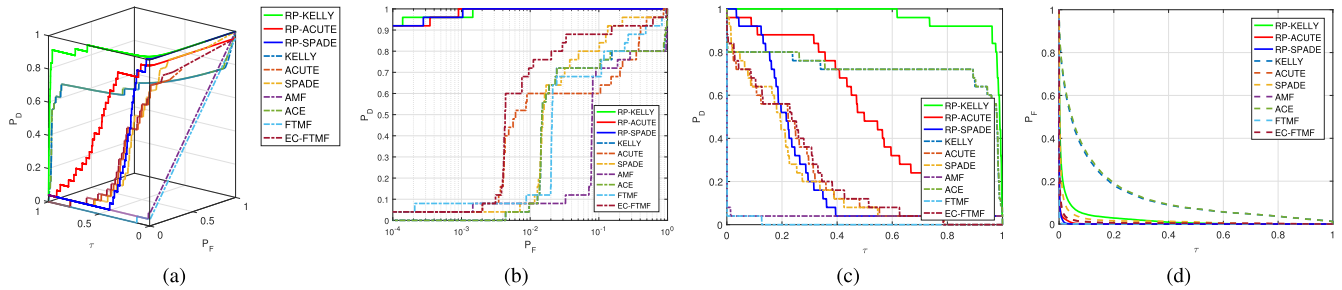

Fig. 12. ROC curves under sample size $n = 264$ on Salinas synthetic image with linear mixing. (a) 3-D ROC curve, (b) 2-D ROC curve of (P_D, P_F), (c) 2-D ROC curve of (P_D, τ), and (d) 2-D ROC curve of (P_F, τ).

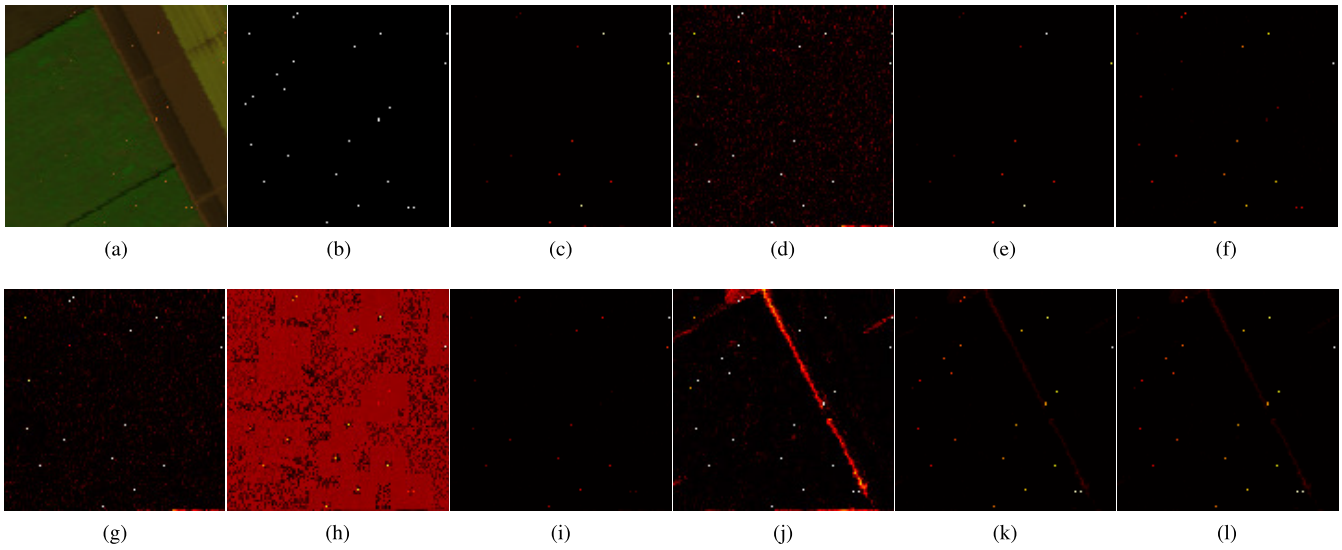

Fig. 13. Detection of performance on Salinas synthetic image with linear mixing. (a) False-color image of synthetic scene with linear mixing, (b) ground-truth map of synthetic scene with linear mixing, (c) AMF, (d) ACE, (e) FTMF, (f) EC-FTMF, (g) KELLY, (h) ACUTE, (i) SPADE, (j) RP-KELLY, (k) RP-ACUTE, and (l) RP-SPADE.

TABLE XI
AUCs ON SALINAS SYNTHETIC IMAGE WITH NONLINEAR MIXING

Performance	CEM	HCEM	ECEM	KCEM	KTCIMF	CTTD	CGSAL	RP-KELLY	RP-ACUTE	RP-SPADE
$AUC_{(D,F)}$	**0.9999**	**0.9999**	**0.9999**	**0.9999**	**0.9999**	0.9340	0.9990	**0.9999**	**0.9999**	**0.9999**
$AUC_{(D,\tau)}$	0.4653	0.8835	0.4224	0.8129	0.8129	0.1559	0.5115	**0.9382**	0.5012	0.5035
$AUC_{(F,\tau)}$	0.0617	**0.0050**	0.0059	0.0821	0.0821	0.0234	0.1406	0.0303	0.0076	0.0102
AUC_{TD}	1.4652	1.8834	1.4223	1.8128	1.8128	1.0899	1.5105	**1.9381**	1.5011	1.5034
AUC_{BS}	0.9382	**0.9949**	0.9940	0.9178	0.9178	0.9106	0.8584	0.9696	0.9923	0.9897
AUC_{TDBS}	0.4036	0.8785	0.4165	0.7308	0.7308	0.1325	0.3709	**0.9079**	0.4936	0.4933
AUC_{SNPR}	7.5413	**176.7000**	71.5900	9.9013	9.9013	6.6624	3.6380	30.9637	65.9474	49.3627
AUC_{ODP}	1.4035	1.8784	1.4164	1.7307	1.7307	1.0665	1.3699	**1.9078**	1.4935	1.4932
Time(s)	**0.1062**	0.4758	13.9176	632.0737	636.8751	0.2063	6.0312	77.7547	78.3372	80.9152

and subsidiary windows are 10×10 and 14×14, respectively, and the sample size $n = 100$. Table XIII reports the AUCs for different projection dimensions. As we observed, the AUCs of the detectors vary considerably for different projection dimensions, indicating that the performance of the detector is sensitive to the choice of projection dimension. When the projection dimension is $k = n/2$, the detection performance of the RP-based detectors consistently outperforms the other cases. Deviations from such ratio of k to n result in a reduced AUC, indicating a decline in detector performance.

G. Evaluation of the SSDW Strategy

The estimation of the statistical properties of the background directly affects the performance of detectors. Therefore, devising an effective strategy for background sample selection is of paramount importance. To this end, we compare the advantages of the SSDW strategy using the Salinas synthetic image with linear mixing. The window sizes for this comparison are consistent with those used in the HyMap Image experiment (as specified in Table II). We denote detectors employing the SSDW strategy under three observation models as SRP-KELLY, SRP-ACUTE, and SRP-SPADE. Conversely, detectors using the general sliding dual-window strategy are named GRP-KELLY, GRP-ACUTE, and GRP-SPADE. Table XIV presents the AUCs for these detectors under both strategies across various sample sizes. Additionally, Fig. 16 demonstrates the 3-D ROC curve and corresponding 2-D ROC curves under sample size $n = 264$.

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TABLE XII
EFFECTIVENESS EVALUATION OF THE RP METHOD

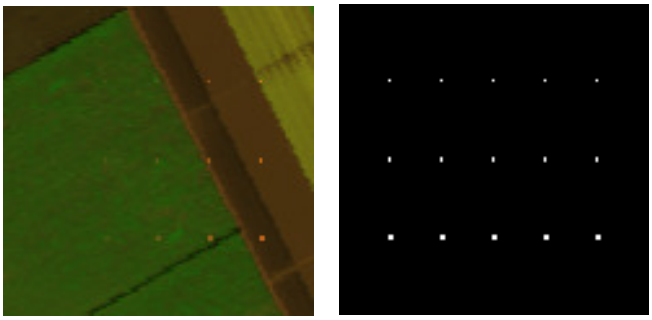
Image	n	KELLY		ACUTE		SPADE	
		RP	no RP	RP	no RP	RP	no RP
SDA-I	200	0.5607	0.5408	0.9627	0.7810	0.9597	0.6869
	264	0.5514	0.5021	0.9295	0.8792	0.9283	0.8584
	336	0.5823	0.5671	0.8525	0.8518	0.8576	0.8391
	416	0.5913	0.5911	0.7506	0.7507	0.7996	0.7988
	504	0.6172	0.6163	0.6515	0.6516	0.7369	0.7370
Salinas synthetic image with linear mixing	144	0.9999	0.9988	0.9999	0.9997	0.9999	0.9993
	200	0.9999	0.9968	0.9999	0.9993	0.9999	0.9987
	264	0.9999	0.9882	0.9999	0.9945	0.9999	0.9876
	336	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
	416	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999

TABLE XIII
AUCS FOR DIFFERENT PROJECTION DIMENSIONS k

Target	k/n	RP-KELLY	RP-ACUTE	RP-SPADE
V3	0.1	0.4316	0.4399	0.7046
	0.3	0.6115	0.6698	0.7120
	0.5	**0.8694**	**0.8342**	**0.7658**
	0.7	0.7186	0.8079	0.6496
	0.9	0.6339	0.6713	0.6626
F2	0.1	0.6419	0.7822	0.3205
	0.3	0.7902	0.8599	0.7223
	0.5	**0.8369**	**0.9161**	**0.7983**
	0.7	0.7636	0.8957	0.6945
	0.9	0.7050	0.7763	0.6998

TABLE XIV
AUCS OF THE SSDW AND GDW STRATEGIES FOR SALINAS SYNTHETIC IMAGE WITH LINEAR MIXING

n	RP-KELLY		RP-ACUTE		RP-SPADE	
	SSDW	GDW	SSDW	GDW	SSDW	GDW
144	0.9999	0.9390	0.9999	0.9888	0.9999	0.9795
200	0.9999	0.9368	0.9999	0.9922	0.9999	0.9864
264	0.9999	0.9667	0.9999	0.9997	0.9999	0.9988
336	0.9999	0.9681	0.9999	0.9993	0.9999	0.9969
416	0.9999	0.9748	0.9999	0.9986	0.9999	0.9944



(a) (b)

Fig. 14. Salinas synthetic dataset with nonlinear mixing. (a) False-color image of the synthetic scene with nonlinear mixing and (b) ground-truth map of the synthetic scene with nonlinear mixing.

The results, in terms of both AUCs and ROC curves, consistently demonstrate the superior performance of detectors employing the SSDW strategy over those using the GDW strategy. Notably, the SSDW strategy-based detectors yielded the best results across different sample sizes. This indicates that the SSDW strategy significantly enhances detector efficacy. The key advantage of the SSDW strategy lies in its ability to select background samples with lower similarity to the target. This approach effectively accentuates the target, thereby improving the detectors' ability to identify it.

IV. CONCLUSION

In this work, three RP-based detectors are proposed for sub-pixel target detection in hyperspectral images. A common issue in local detection is the singularity of the sample covariance matrix when the background sample size is small compared to the spectral dimension. To address this, we derive GLRs for three different observation models based on the RP method under t-distribution background. These detectors significantly enhance detection capabilities while reducing processing time. Furthermore, we develop an SSDW strategy for more accurate extraction of background statistical information. This strategy outperforms the conventional sliding dual-window method, leading to markedly improved detection performance. In experiments using both real and synthetic data, our proposed detectors, especially RP-ACUTE and RP-SPADE, have shown superior performance. Future research will focus on rapidly determining the most suitable projection dimension, a process that currently relies on the assumption of independence. In addition, some ideas and techniques in anomaly detection or unmixing can be employed to further develop more advanced target detectors.

APPENDIX

A. Proof of Theorem 1

Noting that Σ is a positive definite matrix and $\mathbf{P}$ is row-full rank with probability 1, we have that $\mathbf{P}\Sigma\mathbf{P}^{\mathrm{T}}$ is non-singular with probability 1, conditional on the given $\mathbf{P}$.

In the hypothesis testing problem (3), the probability density functions of $[\mathbf{Py}, \mathbf{PZ}]$ under the null and alternative hypotheses, respectively, are

$$p_0(\mathbf{Py}, \mathbf{PZ}) = C_1 |\mathbf{P}\Sigma\mathbf{P}^{\mathrm{T}}|^{-\frac{n+1}{2}} \times \left| \mathbf{I}_k + \frac{\left(\mathbf{P}\Sigma\mathbf{P}^{\mathrm{T}}\right)^{-1}}{\nu - 2} \mathbf{P}\left[\mathbf{y} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right] \times \left[\mathbf{y} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right]^{\mathrm{T}} \mathbf{P}^{\mathrm{T}} \right|^{-\frac{\nu+n+k}{2}} \quad (21)$$

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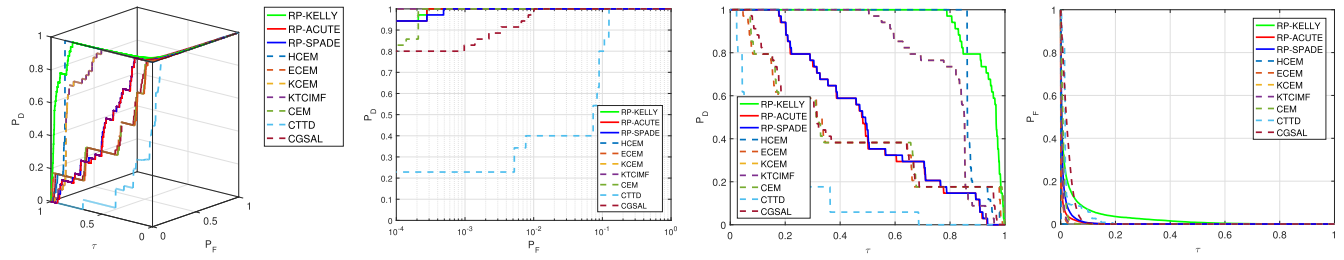

Fig. 15. ROC curves on Salinas synthetic image with nonlinear mixing. (a) 3-D ROC curve, (b) 2-D ROC curve of (P_D, P_F), (c) 2-D ROC curve of (P_D, τ), and (d) 2-D ROC curve of (P_F, τ).

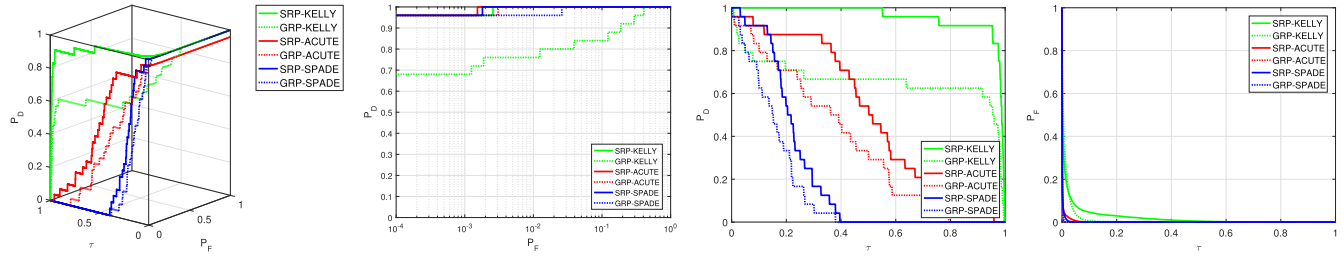

Fig. 16. ROC curves for the two strategies under sample size $n = 264$ on Salinas synthetic image with nonlinear mixing. (a) 3-D ROC curve, (b) 2-D ROC curve of (P_D, P_F), (c) 2-D ROC curve of (P_D, τ), and (d) 2-D ROC curve of (P_F, τ).

and

$$p_1(\mathbf{Py}, \mathbf{PZ}) = C_1 \beta^{-k} |\mathbf{P}\Sigma\mathbf{P}^{\mathrm{T}}|^{-\frac{n+1}{2}} \times \left| \mathbf{I}_k + \frac{\left(\mathbf{P}\Sigma\mathbf{P}^{\mathrm{T}}\right)^{-1}}{\nu - 2} \mathbf{P}\left[\tilde{\mathbf{y}} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right] \times \left[\tilde{\mathbf{y}} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right]^{\mathrm{T}} \mathbf{P}^{\mathrm{T}} \right|^{-\frac{\nu+n+k}{2}} \tag{22}$$

where $\tilde{\mathbf{y}} = (\mathbf{y} - \alpha\mathbf{t})/\beta$ and

$$C_1 = \frac{\Gamma_k((\nu + n + k)/2)}{\pi^{k(n+1)/2}\Gamma_k((\nu + k - 1)/2)}. \tag{23}$$

First, we optimally estimate Σ for fixed $\boldsymbol{\mu}$. Consider the following matrix function:

$$f(\mathbf{X}) = |\mathbf{X}|^{-\frac{n+1}{2}} |\mathbf{I}_k + \frac{1}{\nu - 2}\mathbf{X}^{-1}\mathbf{W}|^{-\frac{\nu+n+k}{2}} = |\mathbf{X}|^{\frac{\nu+k-1}{2}} |\mathbf{X} + \frac{1}{\nu - 2}\mathbf{W}|^{-\frac{\nu+n+k}{2}} \tag{24}$$

where $\mathbf{X}$ and $\mathbf{W}$ are $k \times k$ positive definite matrices. Taking the derivative of $\ln f(\mathbf{X})$ with respective to $\mathbf{X}$ as $\mathbf{0}$, we have

$$\frac{\nu + k - 1}{2}\mathbf{X}^{-1} - \frac{\nu + n + k}{2}\left(\mathbf{X} + \frac{1}{\nu - 2}\mathbf{W}\right)^{-1} = \mathbf{0}. \tag{25}$$

As a result, $f(\mathbf{X})$ attains its maximum at

$$\mathbf{X}_* = \frac{(\nu + k - 1)}{(\nu - 2)(n + 1)}\mathbf{W} =: \tau\mathbf{W}. \tag{26}$$

Conditioning on the given matrix $\mathbf{P}$, by considering $\mathbf{P}\Sigma\mathbf{P}^{\mathrm{T}}$ as a variate of the matrix function $f(\mathbf{X})$, we can obtain the maxima of the likelihood functions under the null and alternative hypotheses are, respectively,

$$l_0(\mathbf{Py}, \mathbf{PZ}) = C_2 \left|\mathbf{P}\left[\mathbf{y} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right] \times \left[\mathbf{y} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right]^{\mathrm{T}} \mathbf{P}^{\mathrm{T}}\right|^{-\frac{n+1}{2}} \tag{27}$$

and

$$l_1(\mathbf{Py}, \mathbf{PZ}) = C_2 \beta^{-k} \left|\mathbf{P}\left[\tilde{\mathbf{y}} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right] \times \left[\tilde{\mathbf{y}} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right]^{\mathrm{T}} \mathbf{P}^{\mathrm{T}}\right|^{-\frac{n+1}{2}} \tag{28}$$

where $C_2 = C_1 \tau^{-k(n+1)/2}(1 + (\nu - 2)^{-1}\tau^{-1})^{-k(\nu+n+k)/2}$.

Second, we derive the optimal $\boldsymbol{\mu}$ for the case of Σ taking its optimal value. Denote

$$\mathbf{M}(\boldsymbol{\mu}) = \mathbf{P}\left[\mathbf{y} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right]\left[\mathbf{y} - \boldsymbol{\mu}, \mathbf{Z} - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\right]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}. \tag{29}$$

By completing the square of $\mathbf{M}(\boldsymbol{\mu})$ with respect to $\boldsymbol{\mu}$, we have

$$\begin{aligned}\mathbf{M}(\boldsymbol{\mu}) &= \mathbf{P}\big(\mathbf{y}\mathbf{y}^{\mathrm{T}} - \boldsymbol{\mu}\mathbf{y} - \mathbf{y}\boldsymbol{\mu} + \boldsymbol{\mu}\boldsymbol{\mu}^{\mathrm{T}} + \mathbf{Z}\mathbf{Z}^{\mathrm{T}} \\ &\quad - \boldsymbol{\mu}\mathbf{1}_n^{\mathrm{T}}\mathbf{Z}^{\mathrm{T}} - \mathbf{Z}^{\mathrm{T}}\mathbf{1}_n\boldsymbol{\mu}^{\mathrm{T}} + n\boldsymbol{\mu}\boldsymbol{\mu}^{\mathrm{T}}\big)\mathbf{P}^{\mathrm{T}} \\ &= (n + 1)\mathbf{P}(\boldsymbol{\mu} - \boldsymbol{\mu}_*)(\boldsymbol{\mu} - \boldsymbol{\mu}_*)^{\mathrm{T}}\mathbf{P}^{\mathrm{T}} \\ &\quad + \mathbf{P}\left[\mathbf{y}, \mathbf{Z}\right]\mathbf{P}_{n+1}^{\perp}\left[\mathbf{y}, \mathbf{Z}\right]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}.\end{aligned} \tag{30}$$

Therefore, $\mathbf{M}(\boldsymbol{\mu})$ attains its maximum at $\boldsymbol{\mu}_* = (\mathbf{y} + \mathbf{Z}\mathbf{1}_n)/(n + 1)$, and

$$\mathbf{M}(\boldsymbol{\mu}_*) = \mathbf{P}\left[\mathbf{y}, \mathbf{Z}\right]\mathbf{P}_{n+1}^{\perp}\left[\mathbf{y}, \mathbf{Z}\right]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}. \tag{31}$$

Thus, the maximum likelihood functions under the null and alternative hypotheses, respectively, become

$$L_0(\mathbf{Py}, \mathbf{PZ}) = C_2 |\mathbf{P}\left[\mathbf{y}, \mathbf{Z}\right]\mathbf{P}_{n+1}^{\perp}\left[\mathbf{y}, \mathbf{Z}\right]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}|^{-\frac{n+1}{2}} \tag{32}$$

and

$$L_1(\mathbf{Py}, \mathbf{PZ}) = \beta^{-k} C_2 |\mathbf{P}\left[\tilde{\mathbf{y}}, \mathbf{Z}\right]\mathbf{P}_{n+1}^{\perp}\left[\tilde{\mathbf{y}}, \mathbf{Z}\right]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}|^{-\frac{n+1}{2}}. \tag{33}$$

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Note that $\mathbf{P}_{n+1}^{\perp} = \mathbf{I}_{n+1} - \mathbf{1}_{n+1}\mathbf{1}_{n+1}^{\mathrm{T}}/(n+1)$ has the following block form:

$$\mathbf{P}_{n+1}^{\perp} = \begin{pmatrix} \mathrm{G}_{11} & \mathbf{G}_{12} \\ \mathbf{G}_{21} & \mathbf{G}_{22} \end{pmatrix} \tag{34}$$

where $\mathrm{G}_{11} = n/(n+1)$, $\mathbf{G}_{12} = -\mathbf{1}_n^{\mathrm{T}}/(n+1)$, $\mathbf{G}_{21} = \mathbf{G}_{12}^{\mathrm{T}}$ and $\mathbf{G}_{22} = \mathbf{I}_n - \mathbf{1}_n\mathbf{1}_n^{\mathrm{T}}/(n+1)$. From $\mathbf{S} = \mathbf{Z}(\mathbf{G}_{22} - \mathbf{G}_{21}\mathrm{G}_{11}^{-1}\mathbf{G}_{12})\mathbf{Z}^{\mathrm{T}}$ and $\bar{\mathbf{z}} = -\mathrm{G}_{11}^{-1}\mathbf{Z}\mathbf{G}_{21}$, we have

$$\begin{aligned}
&|\mathbf{P}[\mathbf{y},\mathbf{Z}]\mathbf{P}_{n+1}^{\perp}[\mathbf{y},\mathbf{Z}]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}| \\
&\quad = |\mathbf{P}[\mathbf{y},\mathbf{Z}]\begin{pmatrix} \mathrm{G}_{11} & \mathbf{G}_{12} \\ \mathbf{G}_{21} & \mathbf{G}_{22} \end{pmatrix}[\mathbf{y},\mathbf{Z}]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}| \\
&\quad = |\mathbf{P}\big(\mathrm{G}_{11}\mathbf{y}\mathbf{y}^{\mathrm{T}} + \mathbf{Z}\mathbf{G}_{21}\mathbf{y}^{\mathrm{T}} + \mathbf{y}\mathbf{G}_{12}\mathbf{Z}^{\mathrm{T}} + \mathbf{Z}\mathbf{G}_{22}\mathbf{Z}^{\mathrm{T}}\big)\mathbf{P}^{\mathrm{T}}| \\
&\quad = |\mathbf{P}\Big(\mathrm{G}_{11}\big(\mathbf{y} + \mathrm{G}_{11}^{-1}\mathbf{Z}\mathbf{G}_{21}\big)\big(\mathbf{y} + \mathrm{G}_{11}^{-1}\mathbf{Z}\mathbf{G}_{21}\big)^{\mathrm{T}} + \mathbf{Z}(\mathbf{G}_{22} \\
&\qquad - \mathbf{G}_{21}\mathrm{G}_{11}^{-1}\mathbf{G}_{12})\mathbf{Z}^{\mathrm{T}}\Big)\mathbf{P}^{\mathrm{T}}| \\
&\quad = |\mathbf{P}\Big(\mathrm{G}_{11}\big(\mathbf{y} - \bar{\mathbf{z}}\big)\big(\mathbf{y} - \bar{\mathbf{z}}\big)^{\mathrm{T}} + \mathbf{S}\Big)\mathbf{P}^{\mathrm{T}}| \\
&\quad = |\mathbf{P}\mathbf{S}\mathbf{P}^{\mathrm{T}}||1 + \mathrm{G}_{11}\big(\mathbf{P}\big(\mathbf{y} - \bar{\mathbf{z}}\big)\big)^{\mathrm{T}}\big(\mathbf{P}\mathbf{S}\mathbf{P}^{\mathrm{T}}\big)^{-1}\big(\mathbf{P}\big(\mathbf{y} - \bar{\mathbf{z}}\big)\big)| \\
&\quad = |\mathbf{P}\mathbf{S}\mathbf{P}^{\mathrm{T}}||1 + \mathrm{G}_{11}\big(\mathbf{P}\check{\mathbf{y}}\big)^{\mathrm{T}}\big(\mathbf{P}\mathbf{S}\mathbf{P}^{\mathrm{T}}\big)^{-1}\big(\mathbf{P}\check{\mathbf{y}}\big)|.
\end{aligned} \tag{35}$$

From (27), (28), and (35), the GLR by applying the RP becomes

$$R(\mathbf{y},\mathbf{Z}) = \frac{\Big(1 + \frac{n}{n+1}\big(\mathbf{P}\check{\mathbf{y}}\big)^{\mathrm{T}}\big(\mathbf{P}\mathbf{S}\mathbf{P}^{\mathrm{T}}\big)^{-1}\big(\mathbf{P}\check{\mathbf{y}}\big)\Big)^{\frac{n+1}{2}}}{\min_{\alpha,\beta}\beta^{k}\Big(1 + \frac{n}{n+1}\big(\mathbf{P}\tilde{\mathbf{z}}\big)^{\mathrm{T}}\big(\mathbf{P}\mathbf{S}\mathbf{P}^{\mathrm{T}}\big)^{-1}\big(\mathbf{P}\tilde{\mathbf{z}}\big)\Big)^{\frac{n+1}{2}}} \tag{36}$$

where $\tilde{\mathbf{z}} = \beta^{-1}(\mathbf{y} - \alpha\mathbf{t}) - \bar{\mathbf{z}}$. Consider the following function:

$$g(\alpha) = \left(\mathbf{P}\left(\frac{\mathbf{y} - \alpha\mathbf{t}}{\beta} - \bar{\mathbf{z}}\right)\right)^{\mathrm{T}}\mathbf{Q}\left(\mathbf{P}\left(\frac{\mathbf{y} - \alpha\mathbf{t}}{\beta} - \bar{\mathbf{z}}\right)\right) \tag{37}$$

where $\mathbf{Q}$ is an arbitrary $k \times k$ symmetric and positive definite matrix. By taking the derivative of $g(\alpha)$ with respect to α as 0, we obtain that $g(\alpha)$ reaches its minimum at

$$\alpha_* = \frac{(\mathbf{P}\mathbf{t})^{\mathrm{T}}\mathbf{Q}\big(\mathbf{P}\mathbf{y} - \beta\mathbf{P}\bar{\mathbf{z}}\big)}{(\mathbf{P}\mathbf{t})^{\mathrm{T}}\mathbf{Q}(\mathbf{P}\mathbf{t})}. \tag{38}$$

For the optimization problem in the denominator of (36), optimizing for α is equivalent to finding the minimum point of $g(\alpha)$ in the case of $\mathbf{Q} = (\mathbf{P}\mathbf{S}\mathbf{P}^{\mathrm{T}})^{-1}$. Hence, α_* can be rewritten as

$$\alpha_* = \frac{\big(\tilde{\mathbf{S}}\tilde{\mathbf{t}}\big)^{\mathrm{T}}\tilde{\mathbf{S}}\mathbf{P}\big(\mathbf{y} - \beta\bar{\mathbf{z}}\big)}{\big(\tilde{\mathbf{S}}\tilde{\mathbf{t}}\big)^{\mathrm{T}}\big(\tilde{\mathbf{S}}\tilde{\mathbf{t}}\big)}. \tag{39}$$

Third, we compute the GLRs for three models. For the additive model, it is merely necessary to minimize α, and its minimum is reached at α_* with $\beta = 1$. Therefore, it immediately yields that the GLR under the additive model has a form (4). For the replacement model, it is sufficient to let $\beta = 1 - \alpha$ in (36) to obtain the GLR (5). For the mixed model, substituting α_* into (36) gives the GLR (6).

B. Proof of Lemma 1

From

$$\begin{aligned}
|\mathbf{B}| &= |\mathbf{A} + \mathbf{P}[\mathbf{y} - \tilde{\mathbf{y}}, \mathbf{0}]\mathbf{P}_{n+1}^{\perp}[\mathbf{y} - \tilde{\mathbf{y}}, \mathbf{0}]^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}| \\
&= |\mathbf{A} + \frac{n}{n+1}\mathbf{P}\big(\mathbf{y} - \tilde{\mathbf{y}}\big)\big(\mathbf{y} - \tilde{\mathbf{y}}\big)^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}| \\
&= |\mathbf{A}|\left(1 + \frac{n}{n+1}\big(\mathbf{P}\big(\mathbf{y} - \tilde{\mathbf{y}}\big)\big)^{\mathrm{T}}\mathbf{A}^{-1}\big(\mathbf{P}\big(\mathbf{y} - \tilde{\mathbf{y}}\big)\big)\right)
\end{aligned} \tag{40}$$

we obtain

$$\begin{aligned}
\frac{|\mathbf{B}|^{\frac{n+1}{2}}}{|\mathbf{A}|^{\frac{n+1}{2}}} &= \left(1 + \frac{n}{n+1}\big(\mathbf{P}\big(\mathbf{y} - \tilde{\mathbf{y}}\big)\big)^{\mathrm{T}}\mathbf{A}^{-1}\big(\mathbf{P}\big(\mathbf{y} - \tilde{\mathbf{y}}\big)\big)\right)^{\frac{n+1}{2}} \\
&= \left(1 + n\big(\mathbf{P}\big(\mathbf{y} - \tilde{\mathbf{y}}\big)\big)^{\mathrm{T}}\Big(\mathbf{P}\widehat{\Sigma}\mathbf{P}^{\mathrm{T}}\Big)^{-1}\big(\mathbf{P}\big(\mathbf{y} - \tilde{\mathbf{y}}\big)\big)\right)^{\frac{n+1}{2}} \\
&= \big(1 + nT^2\big(\tilde{\mathbf{y}}, \mathbf{Z}\big)\big)^{\frac{n+1}{2}}.
\end{aligned} \tag{41}$$

Equation (12) thus holds.

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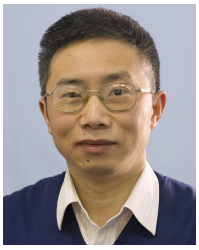

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